Measuring the Task Value of Precision Upgrades in Closed-Loop Agents

Anonymous ACL submission

Abstract

A low-precision agent can call BF16 as a fallback. Such calls can rescue failures and damage successes. Paired branches from the same restored state separate replacement of the current action from takeover of later execution. On Qwen3 agents under simulated quantization in τ^2-bench, replacing the first W4 action that differs in tool structure changes 15.5–20.9% of outcomes. Rescues and damages largely cancel: pooled net gain is $+0.7$ pp (new-task 95% interval $[-2.1, +3.4]$). The pre-registered telecom holdout estimate is negative (-4.6 pp); the protocol classifies it as inconclusive because it omitted a negative-effect category. Lower-bit stress settings also show gains from takeover. Two references test precision attribution: calibration resampling at W4 and tool-argument reordering with an unchanged operation both change outcomes. At shared states, BF16's extra action value over the second checkpoint remains unresolved; equivalence is not established. A frozen uncertainty trigger finds more outcome-changing decisions than random selection without a reliable net-benefit advantage. Evaluating fallback therefore requires measuring the signed task benefit of the intervention that will run, with references that distinguish precision from other sources of change.

1 Introduction

Precision fallback offers a way to intervene when a low-precision agent struggles: call the BF16 model at a selected point in the interaction (Li et al., 2026c). Yet the intervention still has to improve the task. The BF16 model may **replace the current action, take over from the next step, or do both**. Each choice changes a different part of the running trajectory. We ask how much task success each choice gains or loses.

This local question is not answered by comparing complete agents. An end-to-end score measures a model running from the start; a fallback inserts it into a history built by another model. Nor does disagreement reveal the value of the insertion. KL and action differences describe how the models diverge. A changed action can rescue a failure, damage a success, or leave the final outcome intact. The useful quantity is the number of rescues minus the number of damages.

Figure 1a shows a decision from an airline task. Q looks up a reservation id that does not exist; B looks up the user's profile. We execute both actions and continue each with either model. Only BB succeeds. In this case, B's action helps only when B also continues. Evaluating the action alone misses that rescue.

A model that takes over a forked trajectory writes both the next action and the continuation (Gonuguntla, 2026). We measure their contributions separately by running all four action–continuation combinations from the same restored state. We then test an attribution that a flip count leaves open: did the outcome change require higher precision? Another W4 checkpoint (Q′) changes the action without increasing bit width. Reordering tool-argument keys changes the representation without changing the operation. These references give outcome sensitivity a concrete interpretation. We study Qwen3-4B and Qwen3-8B (Qwen Team, 2025) under simulated weight and KV-cache quantization in τ^2-bench (Yao et al., 2024; Barres et al., 2025). Scope checks cover BFCL multi-turn (Patil et al., 2025), ALFWorld (Shridhar et al., 2021), and ordinary sampled states from Mistral-Small-3.2-24B (Mistral AI, 2025) (Appendix F).

Contributions. We contribute a measurement protocol and an empirical test of precision attribution. Building on established step values under a continuing policy (Setlur et al., 2024), the protocol separately evaluates borrowing one action, taking over later execution and changing both. It measures their rescues and damages on shared

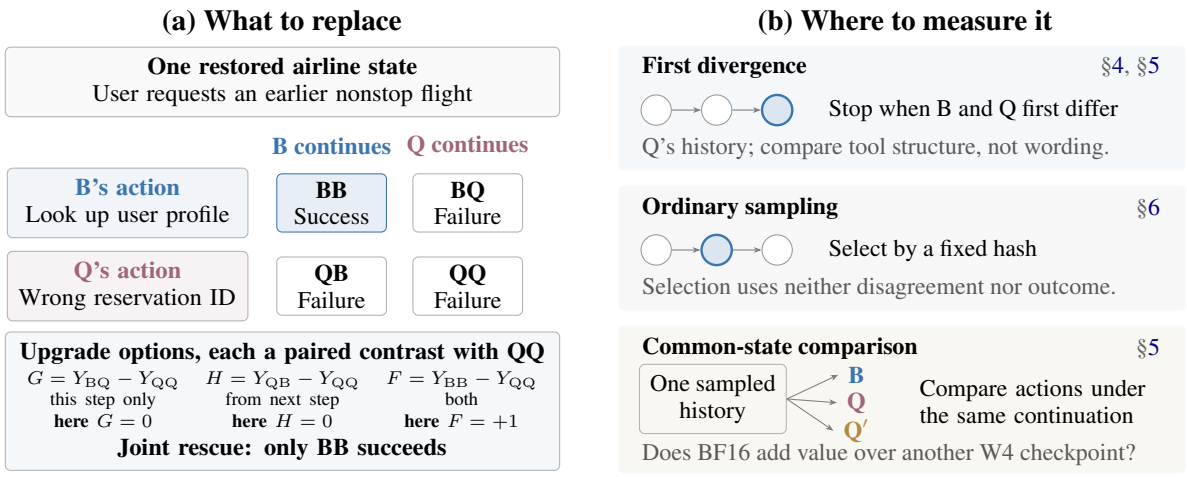


Figure 1: An upgrade has both an intervention and a state population. (a) Cross the current action with the continuing model, then compare each upgrade with QQ. In this airline example (Qwen3-4B, AWQ-style W4; task 31, decision 1 of 12), only joint replacement succeeds. (b) First-divergence states measure consequences at disagreements; ordinary sampling also includes agreements. Comparing multiple action sources on the same sampled history isolates their additional task value. The sampling diagrams are schematic.

states and checks the baseline through exact replay. The empirical study adds two reference interventions: resampling a W4 calibration checkpoint and changing the representation of an unchanged tool operation. A separate shared-state comparison tests BF16's additional action value. Together, they show why a decision's sensitivity to substitution is insufficient evidence that it needs more bits. Unlike independent episodes or a joint switch, the crossed branches identify the value of each upgrade at the same decision. One frozen uncertainty rule provides a test of whether sensitivity can guide beneficial selection.

Three findings build this argument:

1. **A fallback can save tasks and spoil others.** W4 action replacement produces substantial changes in both directions, with little pooled net gain. The action-only estimate is negative on new telecom tasks (§4).

2. **A flip does not identify a precision deficit.** Changing the W4 checkpoint or only the tool-argument order can alter the outcome. The direct shared-state comparison leaves BF16's additional action value unresolved (§5).

3. **Finding sensitive decisions is only part of selection.** Our frozen uncertainty rule finds more outcome changes than random selection, without establishing a net-benefit advantage (§6).

2 Related Work

Quantized agents compared episode by episode. *Flat Score, Amplified Failures* (Jang et al., 2026) is closest in setting: on τ^2-bench retail and telecom, 4-bit agents keep a flat task score while process damage grows, paired tasks flip in both directions, and recovery is equally frequent at both precisions. Its episodes are sampled independently. The failing action and the recovering controller therefore change precision together. We vary them separately at a fixed state. Two-sided flips under compression were first shown on static tasks (Dutta et al., 2024); compression benchmarks compare tasks and models (Dong et al., 2025).

Forks and counterfactual replay. *The Replay Gap* (Gonuguntla, 2026) forks live SWE-bench Verified trajectories for per-step model switching. Its swap replaces the next action and continuation together, corresponding to our joint intervention. Its control holds the model fixed; ours holds the bit width fixed. Replacing one action at a restored state and letting the original policy continue (our "this step only" contrast) locates decisive errors (Zhang et al., 2025), attributes failures (Shah, 2026) and yields training signal (Li et al., 2026a,b); the factorial design itself is standard.

Precision routing, escalation and their signals. DMR (Li et al., 2026c) learns a per-step BF16/W4 router from KL supervision and episode reward; sensitivity has been localised to phases (Lu et al.,

2026), task types (Zhang et al., 2026) and high-entropy, high-KL tokens (Lotfi et al., 2026). Agents also defer single decisions when uncertain (Piatrashyn et al., 2026), learn cost-aware escalation (Lei and Huang, 2026), or are routed per step against execution-verified labels (Yang et al., 2026). Continuing another model's trajectory has its own cost (Ganz et al., 2026). Confidence does not reliably locate consequential steps under executed replay (Zhang, 2026). KL measures disagreement magnitude without identifying its direction (Nikolić et al., 2026). Our evaluation measures action-only, continuation-only and joint substitutions on shared states, using rescue–damage accounting and reference interventions to interpret their task value.

Evaluation noise and user simulators. Prefix caching changed multi-turn tool-use trajectories in 36.2% of episodes at 16-bit and 75.0% at 4-bit (Patodiya, 2026); agentic evaluations vary from run to run (Bjarnason et al., 2026; Chen et al., 2026) and with the user model (Seshadri et al., 2026; Jain and Sandhu, 2026).

3 Measuring Fallback Value

We evaluate a specified fallback at a recorded decision. Every comparison starts from the same history and environment.

3.1 Setting

Tasks and agents. We use τ^2-bench in text mode. The main development panel has 58 retail and 27 airline tasks. We also evaluate 80 telecom development tasks and a holdout of 34 telecom tasks. The holdout protocol was fixed before any agent ran those tasks. The agents are Qwen3-4B-Instruct and Qwen3-8B with thinking disabled; a second family, Mistral-Small-3.2-24B, is a scope check (Appendix F). One greedy Qwen3-30B-A3B-Instruct user simulator serves all agents. Retail and airline success requires a final-database match and normal termination. It excludes some official communication checks. Telecom uses the official task-basis success. BFCL multi-turn and ALFWorld have no LLM user; each item is evaluated once (Appendix A).

Quantization and serving. Q is a simulated (fake-quantized) checkpoint served with BF16 compute on the same path as B. The main W4 settings use GPTQ (Frantar et al., 2023) or AWQ-style scaling (Lin et al., 2024), with group size 128 and a BF16 KV cache. Calibration uses BF16 agent trajectories from 57 retail and airline tasks, disjoint from the development panel. Separate KV8 configurations quantize the cache to INT8 while retaining BF16 weights. Additional quantization settings test the range of upgrade values (Appendix Table 5). B and Q run on vLLM (Kwon et al., 2023) with batch-invariant kernels (He and Thinking Machines Lab, 2025) and deterministic compilation in separate server processes (no shared prefix cache). A configuration is a model with a quantization setting. Each replicate, called a *block*, runs Q over the task pool with a fresh session-identifier line in the system prompt. A *unit* is one task in one block.

3.2 Four branches on one state

Frozen state. Before Q's decision t, we restore state s_t. It contains the full message history, the user simulator's visible conversation, and the used step and error budgets. Replaying the recorded tool calls rebuilds the environment and user-device databases. We save the state before running any branch. Their prompt tokens are identical.

Four branches. Let $Y_{\mathrm{XY}} \in \{0, 1\}$ denote task success when X supplies the current action and Y continues, with X, Y $\in \{\mathrm{B}, \mathrm{Q}\}$. All branches start with the same remaining budget. Each upgrade is compared with QQ: replace this step (G), take over from the next step (H), or do both (F). Figure 1 defines these contrasts. A *rescue* occurs when the upgrade succeeds and QQ fails; a *damage* is the reverse. Their sum is *turnover*, and their difference is *net gain*. These quantities measure task consequences. Additional notation appears in Appendix A.1.

The action being replaced. An action is a complete assistant turn, including its language and structured tool calls. Its message and tool results enter the subsequent history. The continuing model then makes every later decision. Thus BQ asks whether borrowing B's action helps Q finish; QB asks whether B can improve the trajectory after Q's action; BB asks whether changing both helps. The four branches are complete task executions. Each task–block unit receives one intervention. A router that can trigger repeatedly has a different policy value, requiring an end-to-end evaluation.

Reference interventions. We repeat the design with Q′ in place of B. Q′ is another GPTQ-W4 checkpoint of the same model, built by resampling

states and activation rows from the same calibration trajectories. The Q episodes, tasks and prompt blocks stay fixed. This reference tests how much change can occur at fixed bit width. It covers Qwen3-4B and Qwen3-8B GPTQ-W4, with GPTQ-W3 as a stress check. Its protocol was fixed before outcomes were read. Section 5 also compares the action sources at shared states and changes the key order of Q's own tool call.

3.3 Making the counterfactuals auditable

The branches differ in the intended intervention, with history, environment and remaining budgets restored before execution. Outcome-independent no-op audits check that restoring a state preserves its original result: all 924 sampled development states and all 1202 holdout divergent states matched. These checks make the paired comparisons trustworthy under the tested greedy, dense-BF16 serving stack; Appendix A gives the acceptance rules and Appendix C the holdout protocol.

3.4 Sampling and inference

State sources. A first divergent state is the earliest Q-history decision where B and Q differ in action kind, tool calls or normalised arguments. Wording alone does not qualify. Ordinary states are selected by a fixed hash before action generation, one per task and block. A *common state* is an ordinary sampled state at which several action sources are compared; Figure 1b distinguishes the state sources. Full-pool rates include every task–block unit, counting no divergence as zero effect; mode and pattern analyses condition on divergent states. We write G_{all} for the full-pool action-only net gain, and G for its average on divergent states. Contrasts are paired within a configuration. The W4 pool averages four configuration-specific effects equally on the fixed development task set; each configuration supplies its own first divergences. It is not a comparison of configurations at identical decisions. Only complete blocks enter.

Intervals and interpretation. The main figures use new-task intervals, which resample both tasks and prompt blocks, and the holdout uses its registered confidence level. Fixed-pool intervals and the adjusted common-state reading appear in the appendix. An interval containing zero leaves the effect unresolved; equivalence requires a separate test. Captions specify the confidence levels. Each branch measures one realised outcome under the specified greedy stack. A rescue or damage labels that paired execution; its stability across different user or prompt conditions is a separate quantity.

4 Tasks Saved and Tasks Spoiled

A fallback can change many outcomes without improving average success, and action-only replacement can lower success on new tasks. Figure 2 separates rescues, damages and their net benefit.

Rescues and damages nearly balance. We test 85 development tasks over 20 prompt blocks per configuration. B replaces Q's action at their first divergence, then Q continues. Outcomes change in 15.5–20.9% of task–block units (Figure 2a). Counting only these changes would suggest a large effect. Counting their direction gives a different picture: rescues and damages are of similar size. Averaging the four configuration estimates equally, full-pool net gain G_{all} is $+0.7$ pp (new-task 95% $[-2.1, +3.4]$). None of the four configurations establishes a net gain on new tasks. End-to-end model scores appear separately in Appendix A.3.

The telecom holdout has a negative estimate. The pre-registered holdout tests 34 unseen telecom tasks over 12 fresh blocks and three W4 configurations. Here damages exceed rescues (Figure 2b). Full-pool net gain G_{all} is -4.57 pp (crossed 90% $[-7.17, -1.96]$): action-only replacement reduced success in this setting. The protocol nevertheless classifies the result as *inconclusive*, because its frozen readings included positive-gain and small-effect categories but omitted a negative-effect category. Both registered secondary checks passed: rescues and damages were positive, and task fragility repeated across independent prompt blocks. Appendix C gives the readings and secondary tests. Full baseline replay rules out a restoration mismatch at the tested states. The holdout changes domain as well as tasks (Appendix C.1).

This intervention creates a hybrid trajectory: B supplies one action, then Q handles its consequences. The result shows that an action-only fallback can be harmful; it does not identify the cause. For example, difficulty continuing from B's action is one possible explanation, but these data do not isolate it. This differs from running B throughout.

The intervention defines which rescues count. Figure 2c reports all three modes on the W4 and KV8 divergent-state pools. For W4, G, H and F average $+0.5$, $+0.2$ and $+0.7$ pp, respectively; all

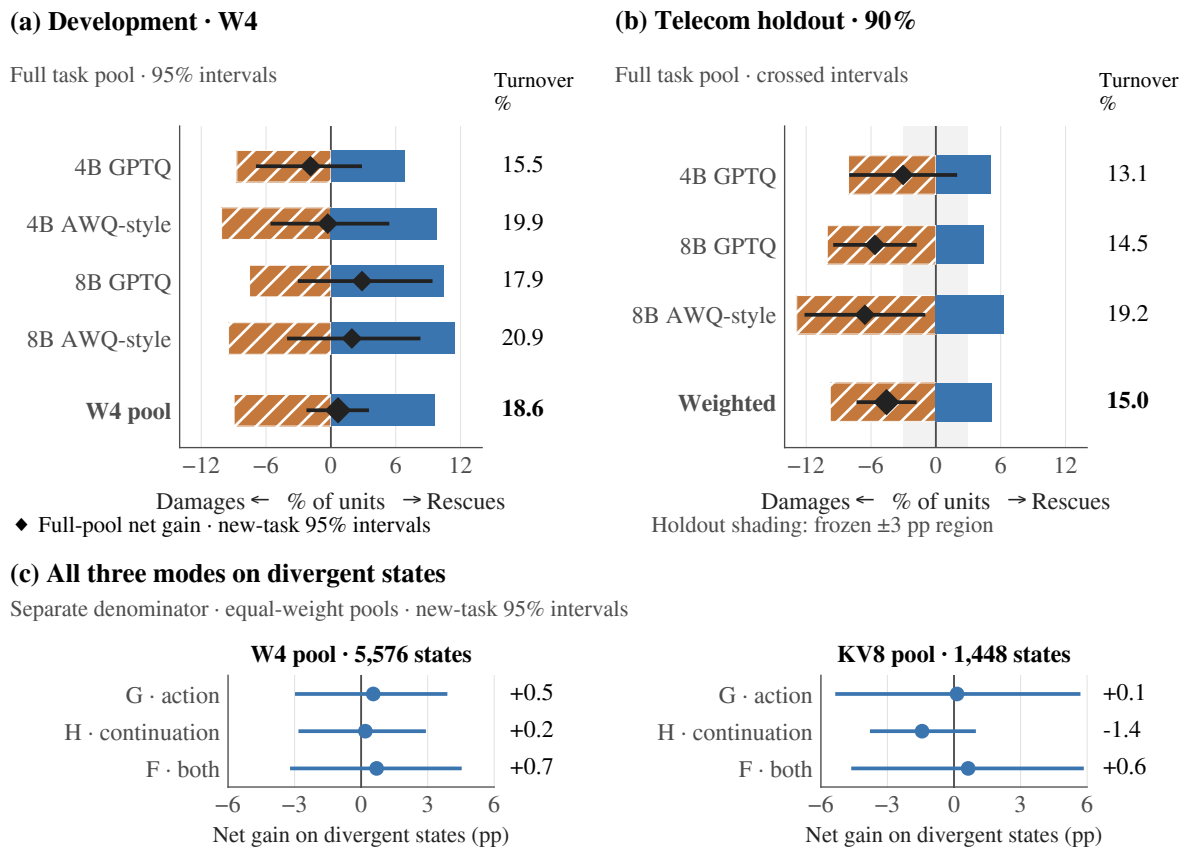


Figure 2: Tasks saved, tasks spoiled and net gain. (a,b) Blue bars show rescues, hatched orange bars damages, and diamonds full-pool net gain G_{all}. Every task–block unit enters, including those without a divergence. (a) Development: new-task 95% intervals. (b) Pre-registered holdout: crossed 90% intervals; shading marks its frozen ± 3 pp region. Bold rows are pooled or primary. (c) The three modes use divergent states only: G replaces the action, H takes over later, and F does both. Their exploratory pooled estimates use new-task 95% intervals. AWQ denotes AWQ-style scaling. Further intervals and pattern analyses: Appendix B.

three new-task intervals cover zero. Similar averages still hide different successes. Joint rescue, where only changing both succeeds, occurs on 4.0% of W4 divergent states. Rescue undone, where B's action succeeds with Q continuing but fails with B continuing, occurs on 4.8%. These are observed shares, not evidence of enrichment over a permutation reference (Appendix B.1). Action-only evaluation misses the former; joint-replacement evaluation misses the latter. Thus a measured gain for one mode cannot stand in for another.

The five lower-bit stress settings provide the complementary case (Table 1). Joint replacement has a positive new-task interval in Qwen3-4B KV4 and Qwen3-8B RTN-W3; the other three remain unresolved. In RTN-W3, borrowing one action has a near-zero estimate while handing over later execution helps. The distinction between modes matters when substantial gains are present as well as when average gains are small.

The practical distinction is between borrowing an action and handing over the task. The former relies on Q to use the substituted action; the latter changes the policy that handles later decisions. Each mode requires its own measurement, even with similar average estimates.

5 Flips Alone Do Not Justify More Bits

Both reference interventions change outcomes without increasing bit width (Figure 3). One changes the checkpoint; the other leaves the tool operation unchanged. They test the inference from a flip to a need for more bits.

Resampling calibration changes outcomes at fixed bit width. Q' resamples decision states and

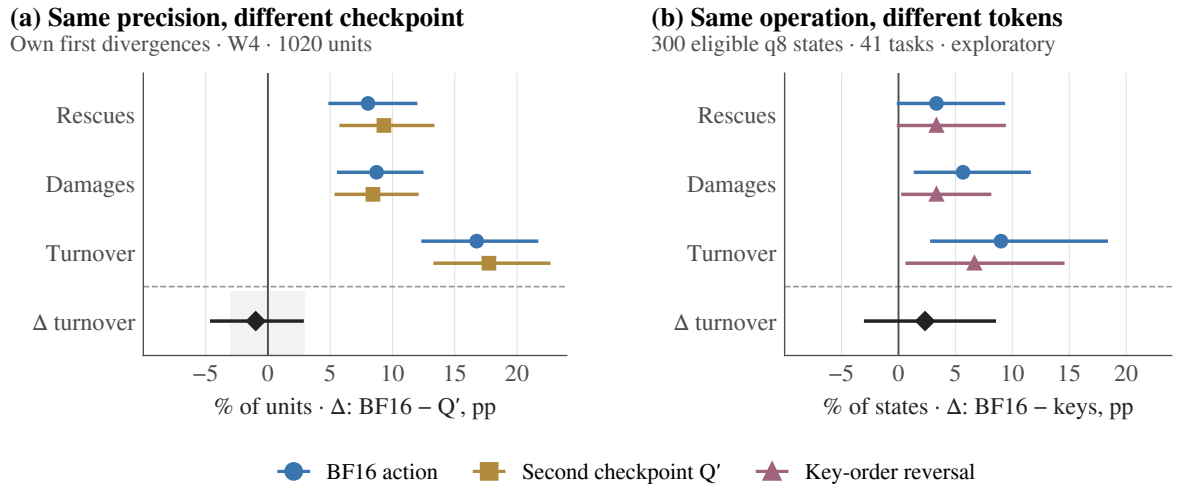


Figure 3: Two reference interventions change outcomes without increasing bit width. (a) A second W4 checkpoint and BF16 are substituted at their own first divergences; lines are new-task 95% intervals. (b) Reversing argument keys preserves the tool operation; lines are the wider of block and new-task 95% intervals on the eligible states. Paired turnover differences are consistently BF16 minus the reference and are separated from rates by a dashed line. The populations differ and the reference rates are not subtracted as a noise floor. The direct shared-state value comparison and all other intervals appear in Appendix D.2 and Table 13.

activation rows from Q's same calibration trajectories. On the matched W4 task–block panel, substituting it produces 17.7% turnover, compared with 16.8% for BF16 (Figure 3a). Both substitutions rescue and damage tasks. Thus large outcome changes also occur when bit width stays fixed. The estimates do not establish equivalence within ±3 pp. The W3 stress comparison also changes outcomes in both directions (Appendix D.1).

The tool operation stays unchanged. At 300 eligible Qwen3-8B states, we reverse the order of keys in Q's tool arguments. Parsing gives exactly the same call, and execution gives exactly the same environment effects. The agent reads a different token sequence afterward. That change alone alters 6.7% of outcomes; replacing the action with BF16's alters 9.0% on the same states (Figure 3b). Their paired difference, BF16 minus key reversal, is +2.3 pp [−2.9, +8.4]. The difference remains unresolved. This exploratory experiment covers one representation change on 41 tasks (Appendix D.4).

Extra precision needs a direct value comparison. The first experiment chooses each substitute's own first divergence. It therefore does not compare B and Q′ at identical decisions. We make that comparison at 510 shared ordinary states from Qwen3-8B BF16 histories, keeping the continuation fixed. BF16's extra action value is +1.0 pp with Q continuing and +0.8 pp with B continuing. Neither adjusted interval establishes an advantage or equivalence. With Q continuing, the interval still allows gains up to +6.2 pp: the study cannot rule out a useful BF16 advantage. Appendix D.3 summarises the estimation precision; Appendix D.2 gives the expansion and reading rules adopted after the first look.

The references support a specific conclusion: outcome sensitivity alone cannot identify a need for higher precision. They expose changes that require neither additional bits nor a different tool operation. They are not a noise floor to subtract from BF16's turnover. Establishing the value of more bits requires a paired benefit comparison on shared states with the same model continuing.

6 Testing Selection for Net Benefit

Our frozen uncertainty rule finds more outcome changes than random selection, without establishing a net-benefit advantage. Figure 4 asks whether the same selected decisions improve task success.

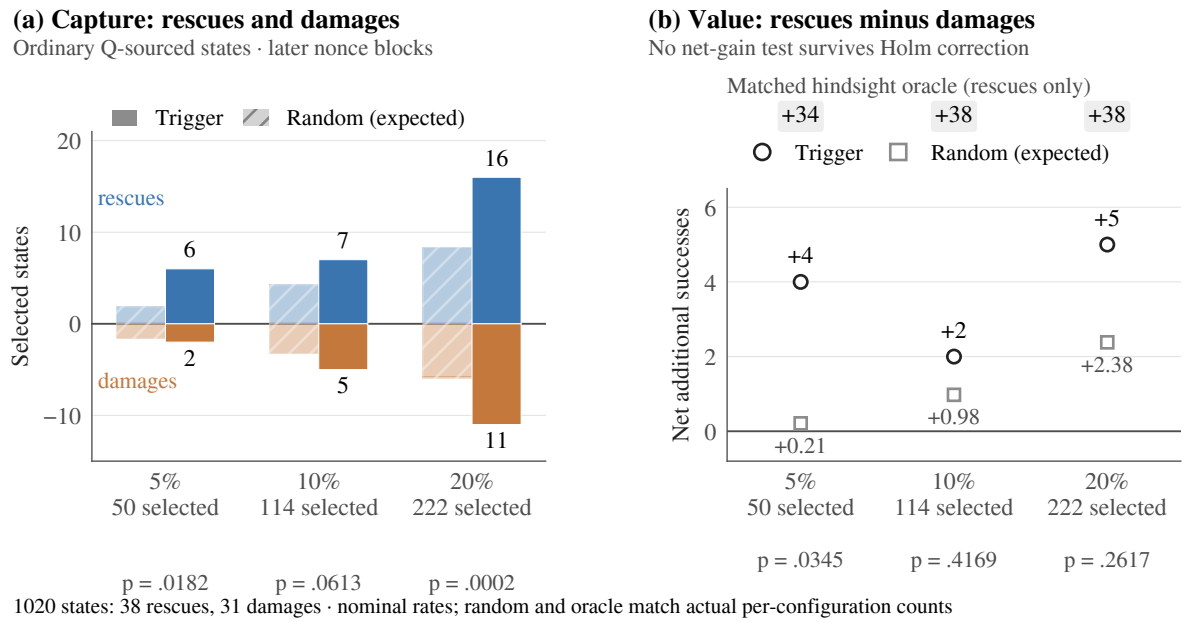


Figure 4: One frozen Q-only margin trigger on 1020 ordinary Q-sourced validation states. (a) Selected rescues and damages versus matched-count random selection. (b) Net additional successes; the grey strip gives the matched-count hindsight oracle, which selects rescues by construction. Nominal rates are shown with actual selected counts. Exact one-sided p-values compare capture in (a) and net gain in (b); no net test survives Holm correction across the three rates. Full values: Table 15.

Choosing before calling BF16. The rule uses only Q's output, so it can choose a decision before spending BF16 computation. We select it on development data, freeze it, and test it on later prompt blocks. On one development block, we choose among entropy, margin and mean log-probability by net gain over random selection. We freeze the winning margin rule before validation: at low minimum token margin, B supplies the action and Q continues. This implements G; margins are known to predict cross-precision divergence (Du et al., 2026). Per-configuration thresholds target 5, 10 and 20% of development states. Validation can select a different fraction, so random and oracle baselines match the actual counts (Figure 4).

We test on two later blocks of the same development tasks, separate from telecom holdout. The states come from Q's own histories. Six configurations (four W4 and two KV8) yield 1020 states, including 69 outcome changes. At the 20% nominal rate, the rule captures 27 changed outcomes against 14.30 expected from matched random selection ($p = .0002$). The other rates and a secondary test on BF16 histories appear in Appendix E.1.

Testing benefit against random selection. The captured states were rescues and damages alike (Figure 4a). None of the three rates shows a reliable net benefit over matched random selection after Holm correction (Figure 4b). No rate was designated primary; exact tests and the nominal 5% result appear in Appendix E.2. The matched hindsight oracle is positive by construction and gives an upper bound (Figure 4b).

On the same selected decisions, the rule succeeds at finding changes without establishing useful selection (cf. Zhang, 2026). This result concerns one rule on later blocks of the same tasks. It motivates a signed-benefit test for learned routing and escalation (Li et al., 2026c; Piatrashyn et al., 2026; Lei and Huang, 2026).

7 Implications for Precision Fallback

A disagreement locates different behaviour; a flip locates a changed task outcome. To evaluate a fallback, measure whether the specific intervention improves success and whether the evidence identifies a precision deficit.

A reference model is not an oracle. BF16 is the full-precision version of the tested model, not a guaranteed better policy. Qwen3-4B GPTQ-W4 reaches 40.0% end-to-end DB success versus BF16's 35.8% (Appendix A.3). Even a model with a better overall score can have unhelpful actions

Stress setting	G	H	F [95% CI]
4B GPTQ-W3	-1.3	$+4.3$	$+1.3$ $[-6.8, +9.4]$
4B KV4-rot	$+5.2$	$+9.4$	$+12.4$ $[+3.0, +21.6]$
8B GPTQ-W3	$+2.1$	$+0.4$	$+2.1$ $[-6.6, +11.8]$
8B KV4-rot	$+3.3$	-0.8	$+4.6$ $[-6.6, +15.9]$
8B RTN-W3	-0.4	$+15.3$	$+18.4$ $[+9.4, +28.2]$

Table 1: All five exploratory stress settings, Qwen3 on retail/airline, three blocks each. Gains use divergent states (pp): action G, continuation H, joint F. Bold F estimates have unadjusted new-task 95% intervals above zero. Full intervals: Table 7.

at particular histories. Inserting those actions also creates trajectories neither model would have generated alone. This explains why a complete-model comparison cannot determine the value of a local fallback; it does not explain the cause of the negative telecom estimate.

Across the five stress configurations, the protocol measures substantial takeover gains in two settings (Table 1). The three modes have different estimates; lower bit width alone does not identify which mode will help. BFCL and ALFWorld extend the comparison to environments without an LLM user; Mistral adds a second model family. These scope checks use separate state populations (Appendix F).

Target the executed upgrade. Borrowing one action asks Q to use it. Handing over the task asks B to handle later decisions. A selector's training target should be the gain of the mode it will execute. Figure 2c shows why similar average estimates do not make the labels interchangeable: a rescue under one mode may fail under another.

Evaluate selection in two stages. Use disagreement or uncertainty to nominate decisions, then test their signed benefit under the intended fallback. Our margin rule passes the capture test without establishing useful allocation. Reporting both makes that distinction explicit.

The selection objective must also retain the sign of the consequence. Turnover gives a rescue and a damage the same positive weight; task benefit gives them opposite signs. Our margin rule illustrates the difference on the same selected decisions. The hindsight oracle establishes an upper bound available after observing the labels, not the ability of Q-visible features to predict them. A learned selector should be tested on held-out decisions against a baseline spending the same fallback budget. For comparisons across modes, that budget should match BF16 computation, because a takeover may use many more decisions than a single borrowed action. We do not measure costs.

From a controlled comparison to expected benefit. Our paired branches measure realised task consequences under fixed execution conditions. Exact replay verifies that the baseline is preserved; it does not make individual rescue and damage labels invariant to other conditions. Key reordering shows that the representation subsequently read by an agent can matter even when the tool operation is unchanged. The shared-state comparison is a separate test of BF16's additional value, and its intervals still allow a useful advantage.

For an expected deployment benefit, the next step is to specify the prompt and user conditions over which the policy should work. Repeat paired branches across that distribution and average their success differences. For stochastic serving, pair the branches under matched nuisance draws. Report average gain separately from how often individual decisions remain rescues or damages. Our intervals account for task and prompt-block sampling, while the current branches use one greedy user simulator and a dense-BF16 serving path. They do not cover changes to the user model or serving kernel. The telecom holdout already shows why new tasks need a direct test: action-only fallback has a negative estimate there. Domain and tasks changed together, so the cause remains unidentified.

8 Conclusion

A precision fallback is valuable when it saves more tasks than it spoils. Our paired branches measure that benefit separately for action replacement, continuation takeover and joint replacement. The W4 panel shows large changes in both directions and little pooled net gain; the telecom holdout has a negative action-only estimate. The exploratory stress panel also shows substantial gains from takeover. Same-precision and unchanged-operation references show why a flip alone cannot diagnose a precision deficit. The frozen selector further demonstrates that locating outcome changes does not establish useful selection. Evaluate each fallback by its signed task benefit under the intended continuation.

Limitations

Inference scope. The contrasts measure realised task effects under the fixed execution stack. Existing intervals leave meaningful BF16 advantages possible (Appendix D.3); we do not infer small effects from a failure to resolve them. Fixed-pool and new-task intervals address different targets; intervals containing zero leave an effect unresolved, while equivalence requires the designated equivalence test. G, H and F are offline intervention values rather than the gain of a deployed selection policy. The rescue-pattern and label-permutation analyses are exploratory and post hoc (Appendix B.1).

Two model families, limited coverage. Most crossed forks use Qwen3 models: 4B and 8B throughout, plus a single Qwen3-14B configuration on BFCL. The second family, Mistral-Small-3.2-24B (GPTQ-W4, RTN-W4), was forked only at ordinary BF16-sourced τ^2-bench states of an 83-task conditional population (Appendix F.3); it has no first-divergent-state forks or same-precision reference. Effects may differ at other scales and in other families. End-to-end runs also exist for Qwen3-14B and Qwen3-32B; apart from that one Qwen3-14B configuration they have no crossed forks, and we draw no conclusion from them (Appendix A.4). A time-boxed Llama-3.1-8B-Instruct check (GPTQ-W4, BFCL multi-turn) was stopped at its gate: BF16 solved 5 of 120 gate items, so the four arms carried almost no outcome variance, and it yields no evidence for that family.

Simulated quantization. Quantized checkpoints are dequantized and served with BF16 compute on a deterministic path. Deployed low-bit kernels change the numerics and add serving variation that we deliberately remove. We make no claims about latency, memory or deployed kernels, and the cost of a BF16 fallback is outside our scope.

One user simulator; two small controls. All τ^2-bench episodes use a single greedy simulator, and continuation effects include its replies. Both second-simulator controls replace only the user after a fixed history, cover 78 and 80 states, and say nothing about real users. The first-divergent-state control has 20 states per configuration, too few to confirm or refute few-pp contrasts. Neither control can be reproduced draw for draw: the replacement user is a hosted Sonnet 5.5 model sampled with its default settings, without temperature or seed control. In the ordinary-state control, Claude Code subagents answered many relay requests in one context, so a responder could see other arms or replicates of the same conversation; the first-divergent-state control removed this exposure by issuing one isolated, single-request call per user turn. The same-precision reference (§5) covers q4 and q8 GPTQ-W4 on the development panel (six blocks) and GPTQ-W3 on three blocks, and swaps one calibration sample for another, re-drawn from the same calibration trajectories; it does not cover checkpoints calibrated on independent datasets. The six-block same-precision panel is the longest complete common prefix under its frozen rule; its seventh q8 block is incomplete. This is separate from the common-state experiment and its extension. It is our only reference for how much turnover and continuation dependence a change of model without a change of bit width produces, and it says nothing about AWQ, KV8, RTN or rotated KV4.

Small confirmation and partial state sets. The pre-registered holdout covers 34 telecom tasks and three configurations. Its pre-registered readings had only positive-gain and small-effect categories; omitting a negative-effect category was a design omission, and it leaves the protocol able to classify the negative estimate only as inconclusive. It applies only to telecom, these three configurations and this simulator. A post hoc replay of all 1202 divergent states reproduced every original QQ suffix (Appendix C); it rules out a replay artifact, not other causes of the negative estimate. Most development results come from retail and airline, so the holdout confounds a domain shift (telecom) with generalisation to new tasks. On 80 telecom development tasks, the two q8 holdout configurations leave the net effect unresolved (Appendix C.1), but their 6-block intervals are wide and q4 has no telecom development forks. These data cannot separate a telecom domain effect from heterogeneity among the 34 holdout tasks. First-divergent-state forks intervene only at the first tool-structure divergence per task and block. Ordinary-state forks sample one state per task and block; their main analyses use three blocks, so their block intervals rest on two degrees of freedom, and the later panels use two to six blocks. Prompt nonces define replicate blocks; they are a researcher-chosen perturbation, not a deployment distribution.

Ethics Statement

This work involves no crowdworkers, no human annotation and no personal data. Where a user is present (τ^2-bench), it is simulated by an LLM; tasks are public benchmark tasks, and all agents are openly released Qwen3, Mistral-Small-3.2 and (gate check only) Llama-3.1 checkpoints, or simulated-quantized versions of them. Outcomes use the automatic task-success endpoints defined in §3.1, including the database-only retail and airline checks, and action categories are assigned by fixed rules; reported values map to committed analysis outputs (Appendix A). The main risk we see is over-reading. A small forked-state gain on one simulator and a fixed task pool does not certify a quantized agent for deployment, and our end-to-end comparisons do not establish equivalence. Deployment decisions in consequential settings should not rest on these measurements alone.

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A Measurement Setup and Replay Checks

The splits, state restoration and replay checks below specify the measurement in §3.

Evidence provenance. Reported values map to committed analysis outputs and frozen manifests. The supplement retains the complete primary comparisons, acceptance checks and protocol amendments needed to assess the paper's claims.

Guide to the supplementary evidence. Appendix A specifies the measurement and its replay checks. Appendix B expands the three upgrade modes, the illustrative rescue patterns and repeatable task fragility. Appendix C documents the independent holdout and full baseline replay. Appendix D supplies the precision-reference comparisons and adaptive-analysis timeline. Appendix E supports the frozen selection rule. Appendix F supplies the environment, model and quantization scope checks discussed in §7. Each section gives the checks needed to interpret its corresponding main-text claim.

Splits. The development split has 58 retail, 27 airline and 80 telecom tasks. The holdout has 34 telecom tasks, and its pre-registration hash is recorded in the split file. The telecom split was fixed from τ^2-bench's 114-task base split by a stratified, salted hash before any run. The external benchmarks use BFCL multi-turn (base, miss_func, miss_param and long_context, 197 items each) and ALFWorld (134 out-of-distribution games). Items were selected by hash in manifests committed before any model run.

Success endpoints. For retail and airline we use the final-database match with normal termination. This is not τ^2-bench's official reward, which additionally scores natural-language or communicate checks. Telecom uses the official deterministic task-basis success. Context overflow and unparseable output count as failures, except for the explicitly amended missing-data rules of the Mistral panel (Appendix F.3). Infrastructure outages are missing rather than failures. The primary analysis excludes incomplete blocks; panel-specific exceptions are stated with their estimates.

Fork mechanics. The forced action is returned by a fork agent for the one step that follows h_t. A tool call is executed by the real environment, and text is answered by the real simulator, which never reuses observations from the original trajectory. τ^2-bench resets step and error counters when it starts from a history. The fork restores them to the values used at t, so the remaining budget equals the original. Every max_steps and too_many_errors episode on the checked blocks is reproduced by the recount.

Replay acceptance details. In the engineering gate, one B→B field, the official reward, was missing from older trajectories and was recomputed from the stored reward information before the verdict. The QQ no-op matched all 924 sampled development states. Ordinary-state checks matched 328/328 sampled states and a separate 135/135 gate set. On the holdout, the initial 135-state sample and the later full replay of all 1202 divergent states matched (Appendix C). Serving in separate processes closes the prefix-cache channel (Patodiya, 2026). The same-precision run's acceptance counts are in Appendix D.

Determinism of the serving stack. Closed-loop determinism pairs, the same configuration on two nodes at different times, reproduced whole episodes token for token for dense BF16 paths at tensor parallelism 1 and 2 with 1 to 16 concurrent requests (e.g. 257/257 and 1520/1520 episodes). Nine arms that two studies ran independently at the same states were byte-identical. We did not test exactness under KV-cache preemption, or on other models, GPUs or vLLM versions. A deployed AWQ kernel path with fp16 dequantization was not replay-exact, and no reported outcome uses it.

User simulator. The serving manifests identify Qwen3-30B-A3B-Instruct as the user simulator for every primary episode. Some archived state files contain a legacy model label; the serving manifests specify the model actually used.

A.1 Internal names and notation

Glossary. The main text uses plain terms; this supplement also uses the internal names of the analysis code. A *first divergent state* is a *Bank P* state. An *ordinary sampled state* is a *Bank U* state; its sub-banks are drawn from Q's own trajectories (U1) or from trajectories shared by all configurations of a model (U2B: BF16 trajectories; U2Q: Q trajectories). A *common state* is an ordinary state of the common-state design (pre-registered at three blocks, extended to six by amendment),

at which B, Q and Q′ each supply the current action. A *cell* is a configuration, named by model (q4: Qwen3-4B; q8: Qwen3-8B; q14: Qwen3-14B; m24: Mistral-Small-3.2-24B) and quantization setting. Nonce blocks carry ids such as fNN (development) and qN (stress and W3 panels); the holdout blocks have their own ids. Branches are written by action source and continuing model (BB, BQ, QB, QQ; KQ is the key-order branch, Appendix D.4). Symbols: r and d are the rescue and damage rates of the action swap, $M = r + d$ is the turnover and $G_{\text{all}} = r - d$ the net gain on the full pool; Δ_{Q}, Δ_{B} and the off-diagonal share are defined below; $I = \Delta_{\text{Q}} - \Delta_{\text{B}}$ is the interaction of action and continuation (Table 7); $D_{cc} = Y_{\text{QB}} - Y_{\text{BB}}$ compares methods on shared ordinary states (Appendix F.5); $K(c)$ is B's action value beyond Q′'s at a common state under continuation c (Appendix D.2); n_{blk} is the number of complete blocks.

Action swap, rescues and damages. The paired effect of the action swap under each continuation is $\Delta_{\text{Q}} = Y_{\text{BQ}} - Y_{\text{QQ}}$ $(= G)$ and $\Delta_{\text{B}} = Y_{\text{BB}} - Y_{\text{QB}}$, each in $\{-1, 0, +1\}$. When $\Delta_{\text{Q}} \neq \Delta_{\text{B}}$ ("off-diagonal"), the swap's value depends on the continuation. For the action swap, a *rescue* is $(Y_{\text{BQ}}, Y_{\text{QQ}}) = (1, 0)$ and a *damage* is $(0, 1)$; with rates r and d, the turnover is $M = r + d$ and the net gain $r - d$. All contrasts measure effects on task success, not which action is correct.

Same-precision notation. M', r', d' and the off-diagonal share are defined likewise with Q′ for B.

Which states. *Bank P*: for each task, block and cell, the first decision at which a_{B} and a_{Q} differ in kind (tool call, text, stop), tool name, call count or order, or normalised arguments; differences in text only are skipped. *Bank U*: one ordinary decision per task, drawn by a hash fixed before generation, in three nonce blocks (six in the extended q8 common-state design, Appendix D.2), from Q's own trajectories (U1) or from trajectories shared by all cells of a model (U2B: BF16; U2Q: Q).

Denominators and inference. We keep three denominators separate. (1) *Full pool*: every task×block unit, units without a Bank P state counting as zero; r, d, M and $G_{\text{all}} = r - d$ use it. (2) *Divergent states*: the Bank P states of a cell; G, H, F and the off-diagonal share use it. (3) *Bank U*: one state per task. Within a cell all branches share each state and all tasks of a block share its nonce, so within-cell contrasts are paired and identified; cross-cell comparisons are descriptive, because each cell selects its own divergences. With n_{blk} complete blocks (only complete blocks enter), the primary *fixed-pool* interval treats the task pool as fixed (block bootstrap if $n_{\text{blk}} \geq 10$, else block-t, a t interval over block means); the *new-task* interval is a crossed task×block bootstrap ("crossed"). We call a contrast *resolved* under an interval when that interval excludes zero and name the interval; "not resolved" means the reported interval or intervals include zero. Where a frozen protocol sets its own reading rule, we state it with the result.

A.2 Analysis design and amendments

Table 2 gathers the timing information relevant to the main claims. Fixing an analysis before new outcomes are read does not erase information already obtained from earlier outcomes. The result sections retain the corresponding interpretation; no amendment below supplies a new confirmation of the W4 benefit estimates.

A.3 Separating local value from end-to-end success

The replacement gains G_{all}, G, H and F are not end-to-end comparisons. Table 3 gives the end-to-end comparison, computed separately on the same checkpoints, the same tasks and the same 20 nonce blocks as the forks.

A.4 Coverage Matrix

Table 4 lists tasks, states and blocks behind the Qwen3 first-divergence and ordinary-state results. Additional shared-history panels appear in Appendix F.5; the six-block common-state comparison appears in Appendix D.2. It counts tasks, states and complete blocks after the strict admission rule. Stress settings are reported separately from the main W4 population.

A.5 Quantization configurations

Table 5 gives the checkpoint settings from metadata and quantization code. Rotated configurations follow QuaRot (Ashkboos et al., 2024) and SpinQuant (Liu et al., 2025); cache experiments include a replay-compatible KIVI variant (Liu et al., 2024) (Appendix F.5). Not recorded: the AWQ-style row-sample size (4096 is the code default) and the KV-cache scale dtype (the hook computes in fp32); Mistral has no second calibration draw.

Comparison	What was fixed before the relevant runs	Outcome-informed steps and interpretation
Core W4 forks	State-restoration, branching and acceptance protocol	Main consequence and pattern summaries are exploratory.
Telecom holdout	Tasks, configurations, weights, blocks, estimator and readings	Primary reading unchanged; both registered secondary checks pass. Full QQ replay is a later audit.
Same-precision reference	Checkpoint construction and paired design	After an interim look, the "similar" reading was replaced by stricter equivalence tests; estimates were unchanged.
Shared ordinary states	Initial three-block design	Extension and two-look rule fixed after the first look; some state outcomes were already known. Both final comparisons remain unresolved.
Key-order reference	Eligibility and analysis frozen before key-order outcomes were read	Exploratory control; the original QQ and BF16 branch outcomes were reused.
Frozen margin rule	Rule and thresholds frozen before validation blocks	Rule selected on one development block; validation uses later blocks of the same tasks.
Mistral scope check	Conditional task mask and phase-extension criteria	Failure-scoring amendment followed phase 1; scope estimates retain that qualification.

Table 2: Timing of designs and readings supporting the main text. Detailed amendments appear with their corresponding results.

Cell	Q / BF16 (%)	Q−BF16 [95% CI]	NI 3 pp	TOST ±3 pp
q4 GPTQ-W4	40.0 / 35.8	$+4.2$ $[-2.4, +10.5]$	pass	fail
q4 AWQ-style W4	36.8 / 35.8	$+0.9$ $[-5.8, +7.5]$	fail	fail
q8 GPTQ-W4	30.1 / 30.4	-0.2 $[-7.7, +6.8]$	fail	fail
q8 AWQ-style W4	31.5 / 30.4	$+1.2$ $[-5.3, +7.6]$	fail	fail
pool, all four		$+1.5$ $[-2.7, +5.6]$	pass	fail

Table 3: End-to-end closed-loop success, quantized minus BF16 (pp). Units are paired on nonce, and the crossed task×block bootstrap is drawn jointly across cells. Non-inferiority (NI) at a 3 pp margin passes for q4 GPTQ-W4 and the all-four pool. Equivalence (TOST at ±3 pp) passes nowhere. These comparisons therefore support neither a difference nor equivalence.

Cell	τ^2-bench retail (Bank P)	τ^2-bench airline (Bank P)	telecom dev (Bank P)	Bank U (retail / airline)	BFCL / ALFWorld
	58 t / 727 st	27 t / 375 st		3 blocks	
q4 GPTQ-W4	20/20 blocks	20/20 blocks	holdout only	393 / 177 rows	788 items · 573 st / –
	58 t / 986 st	27 t / 453 st		3 blocks	
q4 AWQ-style W4	20/20 blocks	20/20 blocks	–	393 / 177 rows	788 items · 642 st / –
	58 t / 412 st	27 t / 181 st		3 blocks	
q4 KV8	22/22 blocks	22/22 blocks	–	393 / 177 rows	– / –
	58 t / 1017 st	27 t / 456 st	80 t / 477 st	3 blocks	
q8 GPTQ-W4	20/20 blocks	20/20 blocks	6/6 blocks ; holdout	393 / 177 rows	788 · 533 st / 134 · 115 st
	58 t / 1087 st	27 t / 475 st	80 t / 480 st	3 blocks	
q8 AWQ-style W4	20/20 blocks	20/20 blocks	6/6 blocks ; holdout	393 / 177 rows	788 · 548 st / 134 · 119 st
	58 t / 585 st	27 t / 270 st		3 blocks	
q8 KV8	22/22 blocks	22/22 blocks	–	393 / 177 rows	– / –
	58 t / 162 st	27 t / 70 st			
q4 GPTQ-W3	3/3 blocks	3/3 blocks	–	–	591 · 540 st† / –
	58 t / 164 st	27 t / 69 st			
q4 KV4-rot	3/3 blocks	3/3 blocks	–	–	– / –
	58 t / 166 st	27 t / 76 st			
q8 GPTQ-W3	3/3 blocks	3/3 blocks	–	–	788 · 696 st / 134 · 130 st
	58 t / 165 st	27 t / 74 st			
q8 KV4-rot	3/3 blocks	3/3 blocks	–	–	– / –
	58 t / 174 st	27 t / 81 st			
q8 RTN-W3	3/3 blocks	3/3 blocks	–	–	– / –
q4 / q8 KV4 (unrotated)	–	–	–	–	788 · 787 / 788 · 788 st; q8 ALFWorld 134 · 134 st
q14 GPTQ-W4	–	–	–	–	788 · 471 st / –

Table 4: Completed crossed-intervention coverage (t = tasks, st = states, b = complete/found blocks; BFCL and ALFWorld: items · states, one run). Bank U: blocks completed and state rows per domain; blocks f01–f03 are analysed (Appendix F.2). "holdout": the cell is in the telecom holdout (Appendix C). Qwen3-32B has end-to-end runs only; Llama-3.1-8B stopped at its fork gate (Limitations); the q14 (Qwen3-14B) BFCL row is a within-family scale check. q4 RTN-W3 has no complete Bank P block and is omitted. The telecom development results (Appendix C.1) use the 6 blocks listed. †The BFCL base category of q4 GPTQ-W3 was dropped by the QQ no-op rule (Appendix F.4). Only completed, admitted panels are counted.

Config. (models)	Method, weights	Group, grid	Act-order	Calibration	KV cache / activations / rotation
GPTQ-W4 Q (q4, q8)	GPTQ, W4; damp 0.01, block 128, true-sequential	128, asym.	yes	draw c1: 57 BF16 agent-trajectory states (39 retail, 18 airline calibration tasks, disjoint from the 85 dev tasks), one state per task, full context (median 6041 / 6310 tok.); 64,000 activation rows (1122–1123 per state); draw seed 20260729	KV BF16, act. BF16, no rotation
GPTQ-W4 Q′ (q4, q8)	as Q	128, asym.	yes	draw c2, seed 20260730: same 57 tasks and trajectories, re-drawn decision state and rows (5/57 states shared with c1); not disjoint	as Q
AWQ-style W4 (q4, q8)	AWQ-style: per-linear scale $\lvert\bar{x}\rvert^{\alpha}$, 20-point α grid, scale folded into weight, then RTN; no clip search	128, asym.	n/a (GPTQ only)	draw c1 as GPTQ-W4 Q; α fit on a 4096-row sample (code default; per-run value not recorded)	as Q
GPTQ-W3^{s} Q / Q′ (q4, q8)	GPTQ, W3	128, asym.	yes	c1 / c2 as W4	as Q
RTN-W3^{s} (q4, q8)	RTN, W3	128, asym.	n/a	none	as Q
KV8 (q4, q8)	BF16 weights	–	–	none	INT8 K and V, asym. min–max per (token, KV head) over head_dim, no clip, no rotation; fake quant in fp32
KV4-rot^{s} (q4, q8)	BF16 weights	–	–	none	INT4 K and V, same grid as KV8; online normalized Hadamard on head_dim: q and k rotated, V rotated and rotated back
QuaRot-W4A4 (q8)	random-Hadamard rotation (seed 0) + GPTQ W4, MSE clip search	per out-channel, sym.	no	WikiText-2 train, 128 × 2048 tok., seed 0	A4: sym. per-token dynamic, clip 0.9, inputs of qkv, o, gate/up, down, online Hadamard (o, down); KV16, per-head q/k Hadamard after RoPE
SpinQuant-W4A4 (q8)	learned R_1, per-layer R_2 (100 steps, lr 1.5, A4 in training) + GPTQ W4 as QuaRot	per out-channel, sym.	no	weights and rotations: WikiText-2, 128 × 2048, seed 0	as QuaRot-W4A4
KIVI-2 / KIVI-4 (q8)	BF16 weights	–	–	none	K per channel, V per token, group 32, asym.; last 128 tokens full precision
Mistral GPTQ-W4 / RTN-W4 (m24)	GPTQ W4 / RTN W4	128, asym.	yes / n/a	GPTQ: draw c1, 57 states, 64,000 rows, seed 20260729 (no c2 draw)	as Q

Table 5: Quantization configuration of every checkpoint configuration, from the checkpoint metadata. AWQ-style W4 is AWQ-style (scale folded into the weight, then RTN, no clip search). Q′ re-draws the decision state and activation rows from the same 57 calibration trajectories as Q. The W4A4 configurations keep a 16-bit KV cache and calibrate on WikiText-2, so a QuaRot-minus-GPTQ contrast also changes the calibration source. One q8 calibration context (c1) is marked truncated. All checkpoints are fake-quantized and stored in BF16; embeddings, `lm_head` and norms stay BF16. ^{s}stress cell.

B Complete Upgrade Effects and Their Interpretation

This appendix supports §4: replacing a W4 agent's action produces both rescues and damages, and a gain measured for one upgrade mode cannot be reused as the gain of another.

Table 6 expands Figure 2a and adds the stress configurations, with both interval types. Table 7 reports G, H, F and $G - H$; Table 8 adds $F - G$, $F - H$ and the outcome-pattern summaries.

B.1 The Two Illustrative Rescue Patterns

Everything in this section is exploratory and post hoc. It uses the four outcomes of each divergent state in the order BB, BQ, QB, QQ; k is the number of successful arms.

Joint rescue and rescue undone against the reference. A label-permutation reference for the two named patterns keeps k for every state and permutes the four outcome labels uniformly: `1000` then has probability $1/4$ at $k = 1$, and rescue undone (`0100` or `0110`) has $1/4$ at $k = 1$ plus $1/6$ at $k = 2$, so the reference shares are $P(k{=}1)/4$ and $P(k{=}1)/4 + P(k{=}2)/6$, computed on the same states and bootstrap draws as the observed shares (Table 9). In the τ^2-bench W4 pool, joint rescue exceeds the reference by $+0.47$ pp on the fixed task pool ([$+0.06$, $+0.87$]), but the new-task interval spans zero ([-0.33, $+1.37$]); rescue undone is less frequent than the reference (-0.99 [-1.58, -0.42]; new tasks [-2.19, $+0.27$]). The excess of joint rescue is concentrated in the q8 cells, and no group shows a new-task excess of rescue undone. Because the reference keeps k and does not model execution, a share near it says that the arrangement is about as common as a random arrangement of the same number of successes, not that the outcomes are noise or that the continuation is irrelevant. The observed shares in the W4 pool, with both intervals, are 4.0% for joint rescue (fixed pool [3.5, 4.5]; new tasks [2.7, 5.3]) and 4.8% for rescue undone ([4.1, 5.4]; [3.2, 6.5]).

B.2 Task fragility

All analyses in this subsection are exploratory and post hoc.

The consequences recur on the same tasks. We split the 20 nonce blocks into two halves by an outcome-independent hash of the block name, ranked tasks by turnover on one half and measured the other half (Appendix Table 10). The 17 most fragile tasks (top 20%) carry 41.0% (crossed [26.6, 55.9]) and 42.3% ([27.0, 57.5]) of the held-out turnover in the two directions, against 20% if fragility did not persist. Their turnover is 38.7%, while the other 68 tasks still turn over 13.7%. Both groups have rescues and damages; neither group's net gain is resolved under the crossed interval. This replicates fragility on other nonces of the same tasks; it does not predict which new tasks are fragile.

Split-half fragility (Table 10). The halves are assigned by an outcome-independent hash of the block name (sha256 of a fixed prefix and the block name, mod 2), written to a file before this post hoc analysis loaded outcomes; fragility is a task's mean turnover over the four W4 cells and the defining half's blocks, and ties at the top-20% threshold are broken by a hash. In the reverse direction (rank on the second half, estimate on the first), the 17 fragile tasks turn over 39.0% (t [34.2, 43.9]) and

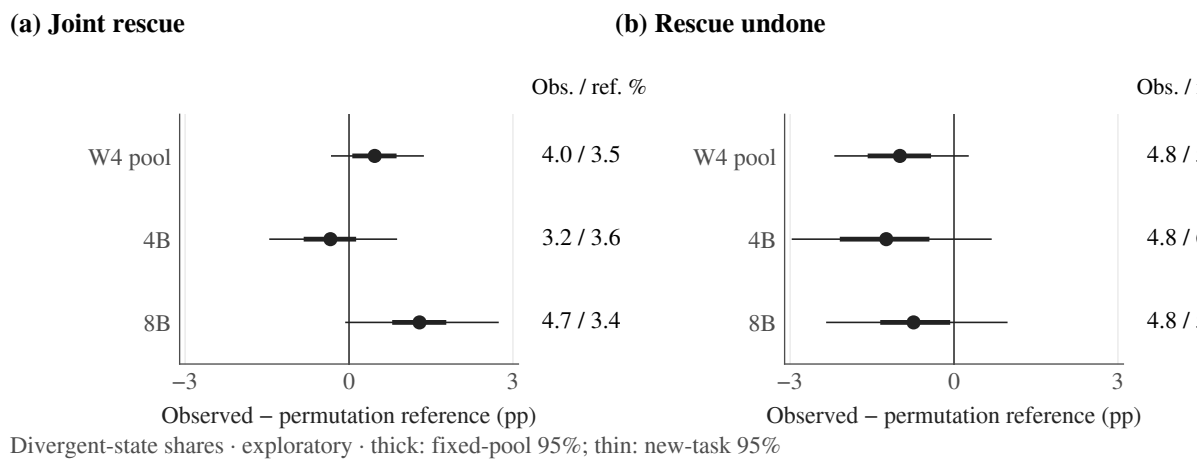


Figure 5: Joint rescue and rescue undone relative to label permutation on W4 divergent states. Thick lines show fixed-pool 95% intervals; thin lines show new-task 95% intervals. The right columns give observed and reference shares. These exploratory comparisons test enrichment, separately from the observed pattern counts in the main text. Values: Table 9.

Cell	units	div.	r	d	M	G_{all} [fixed pool]	[new tasks]
q4 GPTQ-W4	85×20	1102	6.8	8.7	15.5	-1.9 [-3.7, -0.1]	[-6.8, $+2.7$]
q4 AWQ-style W4	85×20	1439	9.8	10.1	19.9	-0.3 [-2.2, $+1.6$]	[-5.4, $+5.2$]
q8 GPTQ-W4	85×20	1473	10.4	7.5	17.9	$+2.9$ [$+1.4$, $+4.4$]	[-2.9, $+9.2$]
q8 AWQ-style W4	85×20	1562	11.4	9.5	20.9	$+1.9$ [-0.6, $+4.3$]	[-3.9, $+8.1$]
W4 pool	85×20		9.6	9.0	18.6	$+0.7$ [-0.3, $+1.6$]	[-2.1, $+3.4$]
q4 KV8	85×22	593	3.1	3.4	6.5	-0.3 [-1.7, $+1.0$]	[-3.0, $+2.3$]
q8 KV8	85×22	855	4.3	3.7	8.1	$+0.6$ [-0.4, $+1.6$]	[-1.9, $+3.0$]
q4 GPTQ-W3^s	85×3	232	9.0	10.2	19.2	-1.2 [-7.0, $+4.7$]	[-8.6, $+5.9$]
q4 KV4-rot^s	85×3	233	11.8	7.1	18.8	$+4.7$ [-4.1, $+13.5$]	[-3.1, $+12.9$]
q8 GPTQ-W3^s	85×3	242	10.2	8.2	18.4	$+2.0$ [-12.5, $+16.4$]	[-6.3, $+11.8$]
q8 KV4-rot^s	85×3	239	11.8	8.6	20.4	$+3.1$ [-14.5, $+20.8$]	[-5.9, $+12.9$]
q8 RTN-W3^s	85×3	255	3.1	3.5	6.7	-0.4 [-3.8, $+3.0$]	[-4.7, $+3.9$]

Table 6: Replacing Q's first tool-structure-divergent action while Q continues, τ^2-bench retail/airline panel (pp). Denominator: the full pool of tasks × complete blocks; units without a Bank P state count as zero. div. = Bank P states. $G_{\text{all}} = r - d$ is a local replacement effect, *not* the end-to-end success difference (Appendix A.3). Fixed pool: block bootstrap (≥10 blocks) or t over block means; new tasks: crossed task×block bootstrap. ^{s}stress cell, 3 blocks.

carry 42.3% of the held-out turnover, the other tasks 13.3%; the fragile tasks' net gain is $+3.7$ pp (t [$+0.6$, $+6.9$]; crossed [-5.0, $+12.4$]) and the rest's $+0.5$. Ranking tasks by BF16 success instead (tertiles on one half) also separates turnover: the hardest tertile turns over 6.7–10.0% and the others 25.4–27.3%, with no net gain resolved under the crossed interval.

Cell	states	G (action)	H (continuation)	F (joint)	$G - H$
		-2.9 [-5.7, -0.2]	$+0.7$ [-1.6, $+3.0$]	-2.5 [-5.3, $+0.2$]	-3.6 [-6.6, -0.6]
q4 GPTQ-W4	1102	[-10.3, $+4.2$]	[-4.0, $+6.0$]	[-9.9, $+4.7$]	[-11.2, $+3.6$]
		-0.3 [-2.6, $+1.9$]	$+0.9$ [-1.1, $+2.9$]	-0.8 [-3.3, $+1.6$]	-1.3 [-3.0, $+0.6$]
q4 AWQ-style W4	1439	[-6.5, $+6.1$]	[-3.7, $+5.5$]	[-7.4, $+5.8$]	[-7.7, $+5.1$]
		$+3.3$ [$+1.6$, $+5.0$]	$+0.4$ [-1.2, $+1.9$]	$+5.4$ [$+3.5$, $+7.2$]	$+2.9$ [$+1.3$, $+4.7$]
q8 GPTQ-W4	1473	[-3.3, $+10.5$]	[-3.8, $+4.9$]	[-2.0, $+13.0$]	[-3.2, $+9.3$]
		$+2.1$ [-0.6, $+4.7$]	-1.3 [-3.5, $+0.9$]	$+0.6$ [-1.5, $+2.7$]	$+3.4$ [$+1.5$, $+5.2$]
q8 AWQ-style W4	1562	[-4.1, $+8.7$]	[-7.1, $+4.6$]	[-6.2, $+7.3$]	[-1.6, $+8.6$]
		$+0.5$ [-0.7, $+1.7$]	$+0.2$ [-1.3, $+1.6$]	$+0.7$ [-0.6, $+2.1$]	$+0.4$ [-0.8, $+1.7$]
W4 pool	5576	[-2.9, $+3.8$]	[-2.7, $+2.8$]	[-3.1, $+4.4$]	[-3.1, $+3.7$]
		-1.0 [-5.3, $+3.2$]	-2.5 [-4.5, -0.5]	-1.2 [-4.7, $+2.3$]	$+1.5$ [-3.1, $+6.0$]
q4 KV8	593	[-9.4, $+7.3$]	[-6.4, $+1.1$]	[-9.3, $+7.7$]	[-7.1, $+9.7$]
		$+1.3$ [-0.9, $+3.4$]	-0.4 [-2.1, $+1.5$]	$+2.5$ [$+0.0$, $+5.1$]	$+1.6$ [-0.6, $+3.9$]
q8 KV8	855	[-4.0, $+6.9$]	[-3.5, $+2.9$]	[-3.5, $+8.1$]	[-3.8, $+7.2$]
		$+0.1$ [-2.8, $+3.0$]	-1.4 [-2.7, -0.1]	$+0.6$ [-1.8, $+3.1$]	$+1.6$ [-1.2, $+4.1$]
KV8 pool	1448	[-5.3, $+5.6$]	[-3.7, $+0.9$]	[-4.5, $+5.8$]	[-3.8, $+6.6$]
		-1.3 [-7.7, $+5.0$]	$+4.3$ [-10.7, $+19.4$]	$+1.3$ [-1.9, $+4.4$]	-5.6 [-15.0, $+3.7$]
q4 GPTQ-W3^s	232	[-9.7, $+6.6$]	[-6.6, $+14.5$]	[-6.8, $+9.4$]	[-15.0, $+4.0$]
		$+5.2$ [-4.5, $+14.9$]	$+9.4$ [$+5.1$, $+13.7$]	$+12.4$ [$+11.0$, $+13.9$]	-4.3 [-18.3, $+9.8$]
q4 KV4-rot^s	233	[-3.4, $+14.2$]	[$+2.1$, $+17.0$]	[$+3.0$, $+21.6$]	[-13.5, $+7.0$]
		$+2.1$ [-13.3, $+17.2$]	$+0.4$ [-7.4, $+8.2$]	$+2.1$ [-7.4, $+11.4$]	$+1.7$ [-9.1, $+12.3$]
q8 GPTQ-W3^s	242	[-6.7, $+12.1$]	[-7.2, $+8.1$]	[-6.6, $+11.8$]	[-7.7, $+11.2$]
		$+3.3$ [-15.4, $+21.9$]	-0.8 [-18.2, $+16.3$]	$+4.6$ [-16.0, $+25.0$]	$+4.2$ [$+2.5$, $+5.8$]
q8 KV4-rot^s	239	[-6.6, $+13.8$]	[-11.2, $+8.5$]	[-6.6, $+15.9$]	[-3.3, $+12.2$]
		-0.4 [-3.8, $+3.0$]	$+15.3$ [$+0.7$, $+29.9$]	$+18.4$ [$+16.7$, $+20.1$]	-15.7 [-27.5, -3.9]
q8 RTN-W3^s	255	[-4.7, $+3.9$]	[$+5.9$, $+25.9$]	[$+9.4$, $+28.2$]	[-24.7, -7.1]

Table 7: Upgrade gains on divergent states (pp). In each two-line cell, the first line gives the estimate and fixed-pool 95% interval; the second gives the new-task crossed 95% interval. G, H and F compare action-only, continuation-only and joint replacement with QQ. Fixed-pool intervals use a block bootstrap over 20–22 blocks, or a t interval over three blocks for ^{s}stress configurations. Pools average within-configuration estimates equally; stress configurations are separate.

Cell	$F - G$	$F - H$	off-diag. %	rev. %	mean I
q4 GPTQ-W4	$+0.5$ $[-1.2, +2.2]$ $[-4.7, +6.1]$	-3.2 $[-6.4, +0.2]$ $[-11.1, +4.4]$	25.0	2.6	$+0.3$
q4 AWQ-style W4	-0.4 $[-2.4, +1.5]$ $[-6.2, +5.4]$	-1.7 $[-3.6, +0.4]$ $[-8.7, +4.7]$	29.3	3.1	$+1.3$
q8 GPTQ-W4	$+2.1$ $[+0.2, +4.0]$ $[-3.0, +7.1]$	$+5.0$ $[+3.2, +6.8]$ $[-2.0, +12.5]$	22.9	2.7	-1.7
q8 AWQ-style W4	-1.5 $[-4.2, +0.8]$ $[-7.6, +4.2]$	$+1.9$ $[-0.1, +3.8]$ $[-3.8, +7.5]$	30.5	4.2	$+0.3$
W4 pool	$+0.2$ $[-1.0, +1.2]$ $[-2.8, +3.1]$	$+0.5$ $[-1.0, +2.1]$ $[-3.5, +4.5]$	26.9	3.2	$+0.0$
q4 KV8	-0.2 $[-3.5, +3.7]$ $[-5.7, +5.3]$	$+1.3$ $[-2.5, +4.9]$ $[-6.7, +9.5]$	15.7	1.2	-2.4
q8 KV8	$+1.2$ $[-0.5, +3.2]$ $[-2.5, +5.0]$	$+2.8$ $[+0.3, +5.5]$ $[-3.1, +8.7]$	14.9	1.2	-1.5
KV8 pool	$+0.5$ $[-1.6, +2.8]$ $[-2.8, +3.9]$	$+2.1$ $[-0.2, +4.2]$ $[-3.0, +7.3]$	15.3	1.2	-1.9
q4 GPTQ-W3^s	$+2.6$ $[-0.6, +5.8]$ $[-5.2, +10.5]$	-3.0 $[-15.1, +9.1]$ $[-12.2, +6.6]$	31.5	3.9	$+1.7$
q4 KV4-rot^s	$+7.3$ $[-2.0, +16.5]$ $[-1.3, +17.1]$	$+3.0$ $[-1.9, +8.0]$ $[-6.2, +12.2]$	33.0	1.7	$+2.1$
q8 GPTQ-W3^s	$+0.0$ $[-11.0, +11.1]$ $[-9.1, +9.1]$	$+1.7$ $[-0.1, +3.4]$ $[-6.8, +11.0]$	23.1	4.5	$+0.4$
q8 KV4-rot^s	$+1.3$ $[-1.8, +4.3]$ $[-7.2, +9.9]$	$+5.4$ $[+0.8, +10.0]$ $[-2.5, +14.6]$	31.0	4.6	-2.1
q8 RTN-W3^s	$+18.8$ $[+13.8, +23.9]$ $[+10.2, +27.8]$	$+3.1$ $[-13.0, +19.2]$ $[-7.1, +13.3]$	29.0	0.4	-3.5

Table 8: Paired mode contrasts and outcome patterns on the states of Table 7. Two-line cells give fixed-pool and new-task 95% intervals in that order. Off-diag.: $\Delta_{\mathrm{Q}} \neq \Delta_{\mathrm{B}}$; rev.: $\Delta_{\mathrm{Q}} = -\Delta_{\mathrm{B}} \neq 0$; interaction $I = \Delta_{\mathrm{Q}} - \Delta_{\mathrm{B}}$. Intervals are unadjusted. The four core W4 $G - H$ tests form a Holm family: q4 GPTQ-W4 has adjusted $p = .069$; all four new-task intervals cover zero. A positive $G - H$ compares two upgrades, not either upgrade against QQ.

A. Joint rescue

Group	states	obs.	ref.	obs.−ref. [fixed pool]	[new tasks]
τ^2-bench W4 pool	5576	3.96	3.48	$+0.47$ [$+0.06$, $+0.87$]	[-0.33, $+1.37$]
q4 W4 cells	2541	3.22	3.56	-0.34 [-0.83, $+0.13$]	[-1.46, $+0.88$]
q8 W4 cells	3035	4.69	3.40	$+1.29$ [$+0.79$, $+1.78$]	[-0.07, $+2.74$]
τ^2-bench KV8 pool	1448	2.99	2.17	$+0.81$ [$+0.10$, $+1.52$]	[-0.51, $+2.19$]
Telecom q8 W4 pool	957	3.24	3.94	-0.70 [-1.46, $+0.05$]	[-2.47, $+1.15$]
Same-precision W4 (Q′ as B)	784	2.27	3.50	-1.23 [-2.47, -0.04]	[-2.88, $+0.51$]
BF16, same blocks	784	4.96	3.53	$+1.43$ [-0.40, $+3.42$]	[-0.90, $+4.26$]
Mistral, own BF16 states^m	464	3.4	2.9	$+0.5$ [-2.4, $+3.5$]	[-2.0, $+3.6$]

B. Rescue undone

Group	states	obs.	ref.	obs.−ref. [fixed pool]	[new tasks]
τ^2-bench W4 pool	5576	4.80	5.78	-0.99 [-1.58, -0.42]	[-2.19, $+0.27$]
q4 W4 cells	2541	4.77	6.01	-1.24 [-2.09, -0.45]	[-2.97, $+0.69$]
q8 W4 cells	3035	4.82	5.56	-0.74 [-1.35, -0.07]	[-2.34, $+0.98$]
τ^2-bench KV8 pool	1448	2.31	4.58	-2.28 [-2.98, -1.58]	[-3.62, -0.98]
Telecom q8 W4 pool	957	4.18	5.13	-0.95 [-1.70, -0.20]	[-2.63, $+0.61$]
Same-precision W4 (Q′ as B)	784	5.77	5.75	$+0.02$ [-2.33, $+2.25$]	[-2.66, $+2.92$]
BF16, same blocks	784	3.82	6.14	-2.32 [-4.46, -0.19]	[-4.68, $+0.17$]
Mistral, own BF16 states^m	464	4.0	4.7	-0.7 [-2.0, $+0.7$]	[-3.0, $+1.9$]

Table 9: Joint rescue and rescue undone against the conditional label-permutation reference (keeps the number k of successful branches per state), % of divergent states; exploratory, post hoc. 95% intervals: fixed pool (block bootstrap for the 20- and 22-block τ^2-bench pools, t over per-block ratios for fewer than 10 blocks: 6, or 5 for Mistral), new tasks (crossed task×block). Same-precision rows each fork at their own first divergence. ^{m}States whose actions differ, on BF16 trajectories (BB is the BF16 original), five blocks; not comparable with Bank P rows.

Tasks		r	d	M	G_{all}
fragile	est.	21.1	17.6	38.7	$+3.4$
(top 20%)	t	[18.6, 23.5]	[14.3, 21.0]	[34.9, 42.6]	[-1.0, $+7.9$]
	crossed	[14.8, 28.1]	[12.4, 23.3]	[32.3, 45.7]	[-7.1, $+13.7$]
rest	est.	6.5	7.2	13.7	-0.7
	t	[5.2, 7.8]	[6.2, 8.3]	[12.5, 14.9]	[-2.7, $+1.2$]
	crossed	[4.4, 9.0]	[4.8, 9.9]	[10.2, 17.5]	[-4.1, $+2.4$]

Table 10: Repeatable task fragility (exploratory, post hoc; four τ^2-bench W4 cells, equal weight). Tasks are ranked by turnover on one hash-defined half of the 20 nonce blocks and measured on the other half (pp of units; 95% intervals, t over held-out blocks and crossed task×block; direction A to B shown). A replication on other nonces of the same tasks, not a prediction for new tasks. Reverse direction: text.

C Telecom Holdout and Full Baseline Replay

This appendix supports the holdout statement of §4: the net estimate is negative, the frozen readings classify it as inconclusive because they had no negative category, and both pre-registered secondary checks passed.

The protocol was frozen before any holdout unit was run, and its hash was recorded in the split manifest. It fixes the following design:

- tasks, cells, weights and nonce blocks;
- two fixed halves of six blocks each;
- an outcome-blind repair rule, with reserve blocks used only for infrastructure repair;
- the decisional crossed interval (two-way ANOVA moments, normal 90%);
- the readings: positive if $L > 0$, above 3 pp if $L > 3$, small if the interval lies inside ± 3 pp, otherwise inconclusive;
- the secondary families S1 (both $r > 0$ and $d > 0$) and S2 (persona-stratified permutation, 10,000 draws, no outcome screen);
- six stratum families, with Holm correction within each.

The analysis script refuses holdout data unless the hash matches. Two task-cell pairs were excluded as structural, and the complete-case analysis is identical.

Full replay of the shared baseline. QQ originally comes from the recorded episode, whereas the other arms are re-executed. An exploratory excess of only-QQ successes in two q8 cells prompted a direct audit. The initial hash sample matched on 135 of 1202 divergent states. A separately frozen audit then replayed all 1202, checking the original checkpoint, serving configuration and numerical canary before each run. Every replay matched the trajectory, prompts, termination, outcome, budget counters, database and environment state. The original and re-executed arms also use the same official success rule. This verifies the QQ baseline; it does not explain why only QQ succeeds on particular tasks.

Structural exclusions. Re-including the two structurally excluded task–cell pairs in all twelve blocks (post hoc sensitivity, frozen estimator) with every unit set to -1 or $+1$ gives point estimates of G_{all} from -6.68 $[-10.35, -3.01]$ to -2.27 $[-5.96, +1.43]$ pp (crossed 90%). This is a range of point estimates: the interval of the most optimistic scenario includes zero. With their observed units kept in the eleven blocks where they exist and only the two missing units set to -1 or $+1$, the estimates are -4.72 to -4.35 pp, each with a crossed 90% interval below zero. The frozen reading is inconclusive in all 36 combinations. Table 11 gives the confirmatory result. Table 12 gives all six stratum families (Holm within family), which are descriptive. Every level estimate is ≤ 0, and the Holm-significant levels are negative: step-budget T2 and T3, mobile-data issues (-9.09 pp) and the fewest-faults tercile (-6.37 pp).

C.1 Telecom development tasks

On 80 telecom development tasks (6 blocks; fixed-pool 95%), the two q8 holdout cells give $G_{\text{all}} = +2.7$ $[-4.7, +10.1]$ (GPTQ-W4) and -0.6 $[-4.9, +3.7]$ pp (AWQ-style W4), against -5.6 and -6.6 on the holdout; q4 has no telecom development forks.

Cell (weight)	units	excl.	G	r	d	ρ_c
q4 GPTQ-W4 (1/2)	396	1	-3.03 [-7.86, $+1.80$]	$+5.05$	$+8.08$	0.668
q8 GPTQ-W4 (1/4)	408	0	-5.64 [-9.33, -1.94]	$+4.41$	$+10.05$	0.374
q8 AWQ-style W4 (1/4)	396	1	-6.57 [-11.98, -1.15]	$+6.31$	$+12.88$	0.644
Weighted (G_{all})			-4.57 [-7.17, -1.96]	$+5.21$	$+9.77$	0.588

Table 11: Telecom holdout, pre-registered analysis (34 tasks $\times$ 12 blocks; pp; crossed 90% intervals). Per-cell rows are descriptive; the weighted row is decisional (G_{all}, reading: inconclusive) and S1/S2 (both pass). Excl.: structurally excluded tasks (decision/trajectory count mismatch at search). Alongside interval for the weighted row, t over block means (not decisional): -4.57 [-6.08, -3.05].

Family	Level	n	G [crossed 90%]	p	Holm p
step budget[a]	T1	138/129/110	-0.32 [-1.57, $+0.94$]	.6771	.6771
	T2	124/126/162	-2.30 [-3.88, -0.72]	.0165	.0495
	T3	123/144/124	-1.94 [-3.35, -0.54]	.0229	.0495
persona	Hard	11	-5.99 [-12.23, $+0.26$]	.1147	.3355
	None	12	-3.65 [-7.42, $+0.13$]	.1118	.3355
	Easy	11	-4.36 [-9.19, $+0.48$]	.1385	.3355
BF16 difficulty	0	20	-2.35 [-4.09, -0.62]	.0257	.0772
	low	11	-4.54 [-8.90, -0.19]	.0860	.1720
	high	3	-18.75 [-38.00, $+0.50$]	.1091	.1720
task length	T1	9	-5.09 [-9.81, -0.38]	.0755	.1511
	T2	8	-0.78 [-4.46, $+2.90$]	.7270	.7270
	T3	17	-6.17 [-10.49, -1.84]	.0189	.0567
issue type	mms	14	-1.15 [-3.33, $+1.04$]	.3880	.4434
	mobile data	11	-9.09 [-14.98, -3.20]	.0111	.0333
	service	9	-4.40 [-10.32, $+1.52$]	.2217	.4434
faults	T1	19	-6.37 [-9.91, -2.83]	.0030	.0091
	T2	10	-1.94 [-7.58, $+3.69$]	.5704	.5704
	T3	5	-2.92 [-5.18, -0.66]	.0337	.0674

Table 12: Telecom holdout strata: the six pre-registered families (descriptive; Holm within each family over its three levels). pp. For task-level families, n is the number of tasks and G is G_{all} over the tasks in the level. [a]Step budget (remaining steps at the divergence) is unit-level: n is the number of divergent units per cell (q4 GPTQ-W4/q8 GPTQ-W4/q8 AWQ-style W4), and G is the level's contribution to G_{all}, so the three levels sum to G_{all}. Tertile cut points were fixed on telecom development tasks.

D Precision References and Common-State Comparisons

The comparisons in §5 distinguish turnover from extra action value. The results below give the same-precision reference, the direct common-state comparison and the operation-preserving key-order control, with their intervals and analysis timelines.

D.1 Second same-bit-width checkpoint

Per model (final analysis set, $n_{\mathrm{blk}} = 6$), the turnover difference is $\Delta M = +0.6$ [-3.1, $+4.3$] pp in q4 and -2.5 [-5.6, $+0.5$] in q8 (fixed pool, 95%), with crossed intervals [-5.5, $+6.9$] and [-7.3, $+2.7$]. Within ± 3 pp, equivalence is shown for no per-model ΔM or ΔG_{all}; the q8 off-diagonal difference ($+0.40$, 90% [-2.12, $+2.92$]) is the only per-model contrast inside the margin. The protocol originally mapped "interval includes 0 and $|\Delta M| \leq 3$ pp" to "similar"; after the interim result had been seen, an amendment replaced that reading by two one-sided tests, which are stricter, and changed no estimate.

Continuation dependence and the reference intervals. On their own divergent states (784 per design), the swap's effect depends on the continuation in 26.2% of Q′'s divergent states (fixed pool [20.7, 31.3]), against 25.0% for B ([20.6, 29.0]); each same-precision rate (rescues, damages, continuation dependence) has a fixed-pool 95% lower bound above zero (Amendment A1.3).

Acceptance: every state in both designs passed Q replay and every branch passed the prefix and environment checks; QQ no-op samples passed 265/265 (precision) and 242/242 (same precision), and no directory was quarantined. At the unit level, Cohen's κ between the action-side flip indicators of the two substitutions is 0.43 in q4 and 0.61 in q8.

W3 stress check (§5; $n_{\mathrm{blk}} = 3$): every state passed Q replay, QQ no-op samples passed 31/31 and 39/39, and no directory was quarantined. All branches passed the prefix and environment checks except one of 774 in q8 same precision: a Q′Q′ branch that ended in an agent-side context overflow, for which the harness records no prefix comparison; its environment hash and first replayed prompt match the frozen state, and by the protocol it counts as a failure, not an exclusion.

Equivalence tests (W4). The difference in turnover is $\Delta M = M - M' = -0.98$ pp (90% CI [-3.26, $+1.30$]). No precision-specific excess was detected, but equivalence within ± 3 pp is not shown either (two one-sided tests, $p = 0.067$), and the same holds for the net gain ($\Delta G_{\mathrm{all}} = -1.57$ [-5.17, $+2.04$], $p = 0.230$) and, descriptively, for the off-diagonal share (-1.210 [-4.09, $+1.67$], $p = 0.133$, as the mean of per-block ratios, which the interval and test use; -1.216 as the ratio of block sums).

Units affected by both, or by only one, substitution. At W4, 107 units flip under both substitutions, 64 only under B and 74 only under the second checkpoint. A within-task block shuffle, which keeps each task's flip propensity, gives

Reference (status)	States (denominator)	Action replacing Q's (continuation)	Turn-over	Rescues / damages	Net	Paired comparison [interval; 95% unless marked]; reading
Second W4 checkpoint Q′	Own first divergences; q4/q8 GPTQ-W4; 1020 task–block units (85 tasks × 6 blocks per model)	B (Q)	16.8	8.0 / 8.7	−0.7	$\Delta M = -1.0$; fixed [−3.9, +1.9]; new [−4.5, +2.7]. Unresolved; equivalence not shown.
		Q′ (Q)	17.7	9.3 / 8.4	+0.9	
Shared ordinary states	510 BF16-sourced q8 states; 85 tasks × 6 blocks; GPTQ-W4 substitutes	B (Q)	7.1	4.1 / 2.9	+1.2	$K(\mathrm{Q}) = +1.0$; two-look 97% t [−4.2, +6.2]; bootstrap [−1.4, +3.5]. Unresolved.
		Q′ (Q)	8.0	4.1 / 3.9	+0.2	
		B (B; vs. QB)	9.0	5.3 / 3.7	+1.6	$K(\mathrm{B}) = +0.8$; two-look 97% t [−2.2, +3.8]; bootstrap [−2.0, +3.7]. Unresolved; equivalence not shown.
		Q′ (B; vs. QB)	8.6	4.7 / 3.9	+0.8	
Key order (exploratory; code frozen before outcomes)	300 q8 GPTQ-W4 first divergent states whose action is one tool call with at least two argument keys, 41 tasks (% of states)	B (Q)	9.0	3.3 / 5.7	−2.3	turnover, B minus reversed keys: +2.3 [−2.9, +8.4]; not resolved; no equivalence margin
		Q, keys reversed (Q)	6.7	3.3 / 3.3	0.0	

Table 13: Reference comparisons behind Figure 3 and the common-state paragraph of §5. Per-arm intervals appear in Appendices D.1 and D.4; unadjusted common-state intervals and the extension timeline appear in Appendix D.2. Each row group has its own state source and denominator; comparisons are paired only within a group, and the references are never subtracted from B's turnover. Turnover, rescues, damages and net compare the replacing action with Q's action under the same continuation: against QQ when Q continues, and against Q's action followed by B (QB) when B continues. Intervals are those each protocol registers: Q′, fixed pool (t over 6 block means) and new tasks (crossed); common states, t over block means and task-cluster bootstrap, a reading needs both, under the two-look (Pocock) reading that the protocol designates confirmatory and, unadjusted, at the frozen levels (Appendix D.2); key order, the wider of block and crossed bootstrap.

76 [69, 84] units flipping under both, so the two substitutions share affected units beyond what flip-prone tasks explain. The overlap lies in action replacement: over the 1020 W4 units, Cohen's κ between the indicators "B's action rescues" and "the second checkpoint's action rescues" is $\kappa_r = 0.60$, and for damages $\kappa_d = 0.54$. Under continuation-only takeover, the affected units are largely substitution-specific: rescue sets under the continuation-only contrast agree with $\kappa = 0.20$, and 70% of continuation-dependent units are continuation-dependent under one substitution only. The difference in action-side net value (ΔG_{all}) comes only from the substitution-specific units and is not resolved (−1.6 pp; fixed pool [−6.2, +3.0], new tasks [−6.5, +3.6]).

Full substitution (W4). Under full substitution B flips more units than Q′ (193 vs. 149; +4.3 [+0.4, +8.3] pp, one of several contrasts); this is a turnover comparison, distinct from the unresolved action-value comparison above.

W3 stress pair. The protocol's complete W3 panel (Q GPTQ-W3, Q′ a second W3 calibration draw; q4 and q8, $n_{\mathrm{blk}} = 3$, blocks q3–q5, 510 units) is reported next to W4, never pooled with it. The same-precision substitution produces 17.3 pp of turnover, the BF16 substitution 18.8 pp; off-diagonal shares are 27.6% and 27.3%. $\Delta M = +1.6$ pp (95% [−0.66, +3.80]) is unresolved: the preregistered ±3 pp equivalence test fails ($p = 0.055$); that the 90% interval lies just above zero does not change this. ΔG_{all} is equivalent within ±3 pp only in the preregistered pooled population (+1.2, $p = 0.017$); per model it is +3.1 (q4) and −0.8 pp (q8). The off-diagonal difference is −0.288 (ratio of block sums) or −0.307 (mean of

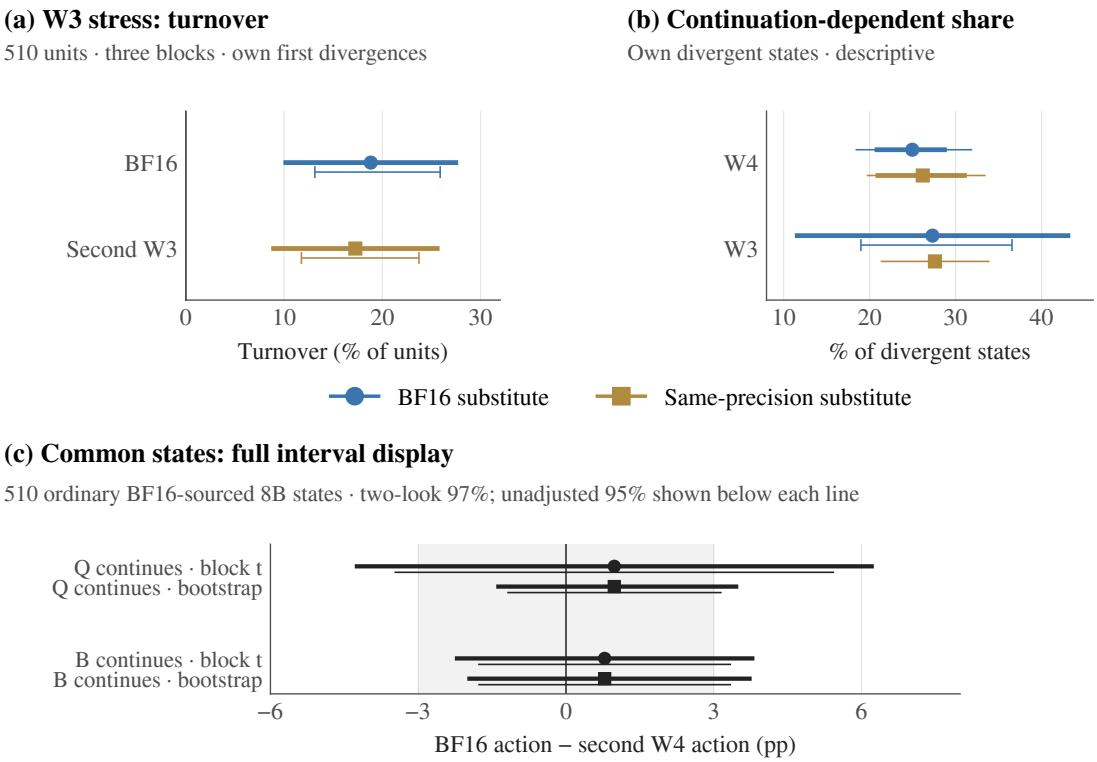


Figure 6: Additional reference results. (a) W3 turnover and (b) continuation-dependent shares: thick fixed-pool and thin new-task 95% intervals. (c) Common-state comparison: block-t and task-bootstrap intervals, with thick two-look 97% and thin unadjusted 95% lines. Shading marks ±3 pp. Each panel has its own state population.

per-block ratios, used by the interval and test: 90% $[-11.0, +10.4]$, $p = 0.27$). On the action side (G), rescued and harmed task×block units overlap ($\kappa = 0.60, 0.68$); the designs may first diverge at different decisions, so this is unit overlap, not a same-state replication. Of 122 units whose G outcome flips, 62 flip under both substitutions, 34 only under BF16 and 26 only under the other W3 checkpoint: 49% respond to only one. Flip volume, affected units and net upgrade value must be measured separately.

D.2 Common states: extra value of the BF16 action

Common-state comparison: details. Above, each design selects its own divergent states. A comparison pre-registered at three blocks removes that difference: on 510 ordinary BF16-sourced q8 states (85 tasks × 6 blocks, sampled by a fixed rule), the current action comes from B, from Q (GPTQ-W4, first checkpoint) or from Q′ (second checkpoint), and B or Q continues. $K(c) = V(\mathrm{B}, c) - V(\mathrm{Q}', c)$ asks what B's action adds beyond another W4 action. A frozen rule claims value only if both intervals exclude zero (Holm), equivalence only if both lie within ±3 pp (two one-sided tests). The design first ran three blocks, and that first look was unresolved (under Q's continuation $K = -0.4$ pp, 95%: t $[-9.3, +8.5]$, bootstrap $[-3.5, +2.7]$; under B's $+2.4$ pp, $[-3.5, +8.2]$, $[-0.8, +5.5]$). Three more blocks were then added. That decision followed the unresolved first look: a dated amendment fixed the extension and the two-look rule after that look and before any extension run had produced output, and the protocol designates this two-look reading (Pocock levels from a normal approximation: about 97% intervals for a difference and 94% for equivalence) as the confirmatory one. The Pocock constant is not calibrated for small-k t intervals, the task bootstrap or the joint rule, so the two-look reading carries no exact error guarantee. Because the extension was chosen after the first look, even the two-look reading cannot fully protect the error rate conditional on the observed first look (protocol D2.2). We therefore print it next to the unadjusted reading at the frozen levels and claim neither an advantage nor equivalence unless both agree. After the fixed extension (six blocks), B's action is not shown to be worth more than Q′'s, and equivalence is not established.

Under Q's continuation, $K = +1.0$ pp (95%: t $[-3.5, +5.4]$, bootstrap $[-1.2, +3.1]$; two-look 97%: $[-4.2, +6.2]$, $[-1.4, +3.5]$). Under B's, $K = +0.8$ pp (95%: $[-1.8, +3.3]$ for both; two-look 97%: $[-2.2, +3.8]$, $[-2.0, +3.7]$). For equivalence under B's continuation, the ordinary 90% intervals lie within ± 3 pp (t $[-1.2, +2.8]$, bootstrap $[-1.4, +2.9]$) and the 94% intervals of the protocol-designated two-look confirmatory reading do not (t $[-1.6, +3.2]$); because the extension followed the first look, we do not claim equivalence. Without the 26 states at which earlier studies had already produced outcomes, K is $+0.5$ and $+0.8$ pp, also not resolved, and the unadjusted reading no longer shows equivalence under B's continuation either; BF16's own outcome at the extension states was also known in aggregate before the amendment (protocol D2.1). Under Q's continuation, B's action flips $M = 7.1\%$ of outcomes (rescue 4.1, damage 2.9) for a net $\Delta = +1.2$ pp (95%: t $[-1.3, +3.6]$); Q′'s flips 8.0% (4.1, 3.9) for $+0.2$ pp ($[-2.3, +2.7]$; within ± 3 pp by two one-sided tests). This auxiliary equivalence compares Q′ with Q, not BF16 with Q′. A second W4 checkpoint's action departs from Q's at least as often as B's does (agreement with Q's action 53.9% against 55.7%; B with Q′ 57.3%). The interaction of K with the continuation is -0.2 pp (95%: t $[-4.5, +4.1]$, bootstrap $[-2.7, +2.4]$; secondary, not resolved). *q4, a separate prospective analysis.* The same design on Qwen3-4B planned three blocks. The frozen quarantine rule removed f01 (below), leaving 170 states. $K = -1.2$ pp under Q's continuation (bootstrap 95% $[-5.9, +3.5]$) and -2.4 pp under B's ($[-6.5, +1.2]$), both not resolved. With two blocks the t interval has one degree of freedom. Under Q's continuation the two block means happen to coincide, so it collapses to a point. That is a coincidence of two values, not a precise measurement or zero variance, and the reading rests on the bootstrap. *Identity checks.* Across both models, 1,238 of 1,240 reference and implicit identity checks passed. Both failures are one q4 state. There the substituted action was semantically identical to the original after parsing, but its token sequence differed in the order of keys inside a tool-call argument. The canonical comparison treats such actions as equal, but the chat template renders the order, so the next prompt differed and the continuation could not be a replay. Every check with a token-identical action passed (869/869). As the frozen rule requires, the block was removed.

Common states, BF16 against Q′, by component. Averaging the paired comparisons equally over B and Q continuations, BF16's extra rescues over Q′ on q8 are $+0.3$ pp and its extra damages -0.6 pp, giving an unresolved mean net difference of $+0.9$ pp. This mean differs from the two continuation-specific values of $K(c)$ above. Each component lies within ± 3 pp under a 90% two one-sided test (exploratory, post hoc; the margin is wide relative to rescue rates of 3–5%). q4 is unresolved.

D.3 Estimation precision

Table 14 summarises the precision of the main unresolved benefit comparisons. The bounds come from their existing reports; they are not minimum detectable effects or a retrospective power calculation. In particular, the shared-state design does not rule out a few-point BF16 benefit. Its 510 rows contain 85 tasks repeated across six blocks, rather than 510 independent observations. Adding blocks and adding tasks address different sources of variation. A prospective power analysis would need to preserve that crossed design and specify the target population, effect size and decision rule.

Benefit comparison	Estimate [interval] (pp)
W4 action-only vs. QQ	$+0.7$ $[-2.1, +3.4]$
BF16 vs. Q′, Q continues	$+1.0$ $[-4.2, +6.2]$
BF16 vs. Q′, B continues	$+0.8$ $[-2.2, +3.8]$

Table 14: Existing interval bounds. W4: full-pool new-task 95%. Shared states: two-look 97% block-t; bootstrap intervals and the joint reading appear above. The populations and confidence levels differ.

D.4 Key-order control

Design. Protocol `KEYORDER_PERTURB_PROTOCOL.md` (ko-v1, exploratory control). Eligible states are first tool-structure divergences of q8 GPTQ-W4 (Bank P) where Q's action is a tool call with at least two top-level argument keys, reversing them leaves the parsed call and the environment effects unchanged, and the next prompt tokens differ (300 of 1473 divergent states, 41 tasks). The arm KQ forces the reversed action and lets Q continue; QQ and BQ are reused from Bank P. Eligibility was fixed without outcomes; the analysis code (sha256-recorded) was frozen before any KQ outcome was read. All four tasks passed acceptance (KQ checks 300/300, shadow 300/300, QQ no-op 35/35, no errors or context overflows).

Two tasks were restarted with resume during the run (Amendment 2026-10-05g).

Results. Conditional on eligible states (pooled ratio; intervals are the wider of block and crossed bootstrap), KQ changes 6.7% [0.8, 14.4] of outcomes and B 9.0% [2.9, 18.2]; KQ has 10 rescues and 10 damages, B 10 and 17, and the flips coincide on 5 rescues and 7 damages. Rescue and damage shares (Figure 3b): KQ 3.3% [0.0, 9.3] and 3.3% [0.4, 8.0]; B 3.3% [0.0, 9.2] and 5.7% [1.5, 11.5]. Over the full 1700-unit pool, KQ contributes 1.2 pp [0.1, 2.8] of turnover against 17.9 pp for the BF16 swap on all divergent states.

E Selecting Decisions for Signed Benefit

This appendix supports §6. It specifies the frozen trigger, matched random and hindsight references, and the development diagnostics used to choose the rule. The net-benefit tests remain unresolved after correction.

E.1 The frozen trigger and its headroom

On the first development block, we scored each ordinarily sampled state at the token level (1140 cell $\times$ state rows, 58 flips), teacher-forcing both models on the generated ids of a_{B} and a_{Q}. We record truncated top-20 entropy, the top-1 margin and a top-20-union KL, not full entropy or KL. All analyses are exploratory, and f01 is development data.

Uncertainty is associated with divergence. Q's entropy at the start of its action separates states where B and Q act differently from states where they act identically, with AUC 0.66–0.91 per sub-bank and model. Over all ordinary states, flips also sit at lower Q margins (U1: AUC 0.31 [0.22, 0.40] for q4, 0.24 [0.13, 0.36] for q8; stratified). Among already divergent U1 states, no Q-visible statistic has an interval excluding AUC 0.5; small-subset associations vary across state sources. This development diagnosis motivated the frozen margin rule, whose validation reuses the tasks with fresh nonce blocks. On the secondary validation population, ordinary states from BF16 trajectories (U2B; f02/f03, 1020 rows, 75 outcome-changing states), the frozen rule captures 4, 11 and 35 outcome-changing states against 3.81, 8.35 and 16.07 expected at random ($p = .5404$ and .1995 at 5 and 10%; the 20% excess is resolved), with net gains of -2, -1 and $+5$ against random $+0.07$, $+1.10$ and $+2.48$.

E.2 Matched references for the frozen trigger

Table 15 compares the trigger of §6 with matched random selection and a hindsight oracle on the same 1020 validation rows (Q-sourced histories, later nonce blocks), containing 38 rescues and 31 damages. The *matched action-replacement oracle* selects, per cell, exactly as many rows as the frozen rule did, taking only rows with $G > 0$ and filling with $G = 0$ rows when a cell has too few rescues; it never selects a damage. It is hindsight selection, not a policy. At the 10% and 20% budgets every rescue fits.

Exact random-selection test. In configuration c, condition on its N_c states, containing R_c rescues, D_c damages and $Z_c = N_c - R_c - D_c$ unchanged outcomes. Draw m_c states uniformly without replacement, matching the trigger's actual count. The random rescue and damage counts have joint mass

$$P_c(r, d) = \frac{\binom{R_c}{r}\binom{D_c}{d}\binom{Z_c}{m_c - r - d}}{\binom{N_c}{m_c}}.$$

We convolve these distributions across configurations. The net statistic is $T = \sum_c (r_c - d_c)$, and its one-sided p-value is $P(T \geq T_{\mathrm{trigger}})$. The capture statistic instead uses $\sum_c (r_c + d_c)$. This accounts jointly for rescues and damages; it is a conditional comparison with random selection on the observed validation rows, not a population-level test for new tasks.

A. Frozen trigger and matched hindsight oracle

Budget	sel.	oracle r/d/net (pp)	frozen r/d/net (pp)	net ratio
5%	50	34/0/+34 (+3.33)	6/2/+4 (+0.39)	4/34
10%	114	38/0/+38 (+3.73)	7/5/+2 (+0.20)	2/38
20%	222	38/0/+38 (+3.73)	16/11/+5 (+0.49)	5/38

B. Frozen trigger against matched-count random selection (Figure 4)

Budget	sel.	net vs. random (p; Holm)	r/d	turnover vs. random (p)
5%	50	+4 vs. +0.21 (.0345; .1035)	6/2	8 vs. 3.48 (.0182)
10%	114	+2 vs. +0.98 (.4169; .5234)	7/5	12 vs. 7.52 (.0613)
20%	222	+5 vs. +2.38 (.2617; .5234)	16/11	27 vs. 14.30 (.0002)

Table 15: Frozen Q-only trigger against a matched action-replacement oracle (hindsight; same per-cell selection counts), on f02/f03 U1 (1020 rows). Net ratio: frozen net over oracle net (not a share of rescues; the frozen rule selects 6, 7 and 16 of the oracle's rescues). Random: matched-count per-cell random-trigger expectation; p: the exact random-selection tail defined below. Holm correction across the three budgets applies to the net tests only.

F Scope Across States, Environments and Models

This appendix supports the scope discussion in §7: ordinary states, environments without an LLM user, a second model family and additional quantization settings. The five stress-mode contrasts in Table 1 are expanded in Table 7. Simulator controls test the sensitivity of the realised outcomes discussed in §7.

F.1 Ordinary states and the user simulator

Ordinary states include agreements. At ordinary states, replacing Q's action changes the outcome less often than at first tool-structure divergences (a descriptive comparison of different state populations, with no interval on the difference). Over blocks f01–f03 (task-equal; t intervals over block means), M per sub-bank is 7.8–9.0% for the q4 W4 cells in U1 and U2B, 4.7–9.0% for the q8 W4 cells and at most 4.7% for KV8. The net gain lies within ± 2.8 pp (U1, U2B) and ± 5.0 pp (U2Q), and every interval covers zero (Appendix F.2). Token-identical actions preserve the outcome under deterministic serving. Reused arms were initially selected by canonical equality and then checked with each model's chat template: 4,113 of 4,114 reuses are token-identical. The remaining pair differs only in JSON key order; its two cross arms were measured separately and reproduced the reused outcomes.

Sensitivity to the user simulator. Text actions act through the user simulator, and changing the simulator moves τ-bench outcomes by 9–15 points (Seshadri et al., 2026; Jain and Sandhu, 2026). We froze a control before running it: the 39 deduplicated text-difference flips with the same tool structure of block f01 and 39 matched non-flip states. Their BQ and QQ branches were each rerun three times with a different user simulator, Sonnet 5.5, answering through a file relay from Claude Code subagents (reproducibility limits in the Limitations). Judged by the sign of the replicate mean, only 7 of the 39 flips keep their direction (95% CI 0.075–0.335). Six reverse, and 26 vanish (15 both succeed, 2 both fail, 9 ties). Among the matched non-flip states, 11 of 39 become flips (0.15–0.45). Replicates of the same arm disagree in 41% of flip arms and in 21–26% of control arms. The text-difference labels did not reproduce reliably under the alternative relay setting. Simulator model, sampling and relay interface changed together, so their separate effects are not identified.

Sensitivity of the four-branch results. Before running, we sampled 80 Bank P states without outcome information (20 per W4 cell, stratified by domain and block) and re-ran all four arms five times each with Sonnet 5.5 as the user, one isolated single-request call per user turn (7896 calls; all integrity checks passed). Per-state arm outcomes track the original single runs (correlation 0.63–0.68). Pooled over cells, $G = -3.0$ $[-10.2, +4.8]$, $H = +5.5$ $[+0.0, +11.2]$ and $G - H = -8.5$ $[-14.8, -2.8]$ pp under Sonnet, against $G - H = -2.5$ $[-11.2, +6.2]$ on the same states under the original simulator (change -6.0 $[-16.2, +4.2]$). With 20 states per cell, this control cannot resolve the few-pp contrasts of Table 7 (on this sample even the original simulator gives $G - H = 0.0$ in both q8 W4 cells), so it neither confirms nor reverses the within-cell orderings.

F.2 Ordinary-state estimates by history source

Table 16 gives the per-state turnover behind Appendix F.1, over blocks f01–f03. Exact action agreement is 0.43–0.83 per cell and sub-bank.

F.3 Models compared on their own ordinary states

Mistral population, scoring and coverage. The Mistral panel applies the same row rule to Mistral's own BF16 trajectories on the 85 retail and airline tasks. Seven BF16 source rows are missing because the source episode ended in an invalid tool call produced by the model itself (airline task 44 in all six blocks, airline task 13 in f01). A

Cell	U1 (n=255)	U2B (n=255)	U2Q (n=60)
q4 GPTQ-W4	.090 (11/12)	.082 (10/11)	.083 (4/1)*
q4 AWQ-style W4	.082 (7/14)	.078 (7/13)	.083 (3/2)
q4 KV8	.047 (4/8)	.039 (5/5)	.017 (0/1)
q8 GPTQ-W4	.047 (8/4)	.067 (9/8)	.050 (3/0)*
q8 AWQ-style W4	.086 (14/8)	.090 (15/8)	.067 (3/1)
q8 KV8	.035 (5/4)	.031 (6/2)	0 (0/0)

Table 16: Bank U, blocks f01–f03: per-state turnover M (rescues/damages), task-equal (one state per task and block; 85/85/20 states per block). G lies within ± 0.028 in U1 and U2B and within ± 0.050 in U2Q, and every interval includes 0. Intervals are t intervals over the three block means and are wide (e.g. q4 GPTQ-W4 U1: .090 [.029, .151]). *GPTQ-W4's U2Q states are its own U1 states, with concordant outcomes on 20/20 duplicates per block; they are counted once in cross-sub-bank totals.

dated amendment, fixed from the BF16 source runs before any Mistral fork output existed, excludes both tasks in every block and cell, leaving 83 tasks. The selection was outcome-blind, but BF16 Mistral fails these tasks structurally, so the estimates describe this conditional population, not the 85-task panel, and are not compared with q8 as if the populations were equal. A branch that ended because the harness could not accept the agent's own message (an empty or invalid tool name, or an empty message) is scored as a failure, as in end-to-end evaluation. This rule was added after the Mistral phase-1 results had been read; without it the Mistral primary comparison is not estimable from phase 1 (one complete block). The analysis without the rule (which leaves such an arm missing and fails its block) and one applying the rule only to data not yet read when it was fixed are reported alongside; both admit the same three blocks for GPTQ − RTN ($+1.2$ [-13.8, $+16.2$]). Coverage: GPTQ-W4 has 498 states on six blocks and RTN-W4 415 states on five. Full estimates on the 83 tasks (GPTQ-W4 on six blocks, RTN-W4 on five; Table 17): in GPTQ-W4, B's action rescues $r = 4.6$ [2.4, 6.8] and damages $d = 5.4$ [2.4, 8.5] pp of states ($M = 10.0$ [7.0, 13.1]; fixed-pool 95%), and no mode's net value is resolved ($G = -0.8$ [-5.2, $+3.6$], $H = +1.4$ [-3.0, $+5.8$], $F = +0.0$ [-4.2, $+4.2$]). RTN-W4 is similar ($M = 10.4$ [6.3, 14.4]). The pre-registered contrast GPTQ − RTN is not resolved: $\Delta D_{cc} = +2.2$ [-3.5, $+7.8$] pp, $p = .35$, on five blocks, two of which are admitted only by a scoring rule adopted after an interim look (above).

F.4 Upgrade effects on BFCL and ALFWorld

Table 18 gives the replacement quantities of §4 (r, d, M and G_{all} over all items, non-divergent items counting as zero) and applies the four-branch analysis to the external benchmarks, from the raw four-arm outcomes on divergent items (one run per item; item bootstrap within benchmark category). The four-branch columns are post hoc and not part of any pre-registered analysis. In the unrotated KV4 configurations, $Y_{\text{QQ}} = 0$ on every state: all positive upgrades are rescues. Action-only replacement remains near zero, whereas continuation and joint upgrades can help. In BFCL q4 GPTQ-W3, the base category was dropped whole by the QQ no-op rule, leaving 591 items and 540 states. Off-diagonal shares are 4.7–17.1% in BFCL W4 cells (pool 10.3%) and 32.8–40.0% in q8 ALFWorld W4 (pool 36.4%). The ALFWorld GPTQ-W3 stress cell has $H = +38.5$ and $G - H = -34.6$ [-43.5, -25.4] pp; the unrotated KV4 rows also distinguish continuation takeover from joint replacement.

F.5 Quantization methods on shared ordinary states

These scope checks ask how the three upgrade values change across quantization settings. Methods share ordinary BF16-sourced states (U2B); the separate Mistral population is specified in Appendix F.3.

Interpretation. The rows below test the range of action-only, continuation-only and joint upgrade values. They report every admitted configuration, including the unquantized path controls. They are descriptive scope checks, not a ranking of quantization algorithms. The configurations differ in calibration pipelines as well as numerical transformations. KIVI is a replay-compatible variant: each decoding step reconstructs a quantized view from the full-precision cache rather than retaining a compressed cache across turns. Admission and

Cell	blocks	states	G	H	F
GPTQ-W4	6	498	-0.8 $[-5.2, +3.6]$ $[-5.8, +4.2]$	$+1.4$ $[-3.0, +5.8]$ $[-4.0, +6.8]$	$+0.0$ $[-4.2, +4.2]$ $[-6.0, +6.0]$
RTN-W4	5	415	$+0.7$ $[-2.7, +4.2]$ $[-3.6, +5.5]$	$+1.9$ $[-3.8, +7.7]$ $[-4.6, +8.0]$	$+2.4$ $[-2.2, +7.0]$ $[-3.9, +8.4]$

Table 17: Mistral-Small-3.2-24B on its fixed 83-task conditional population. pp; fixed-pool 95% t intervals, then crossed task×block 95% intervals. Missing argument-JSON errors leave RTN f04 excluded, whereas model-produced invalid tool names are scored as failures under the dated amendment above. None of the three mode intervals excludes zero.

one-time infrastructure re-run rules are recorded in the supplementary panel protocol.

All admitted ordinary-state upgrade effects. Table 19 reports all three upgrade modes for every admitted configuration, with both fixed-pool and new-task intervals.

A. Full item pool

Cell	items	r	d	M	G_{all}
BFCL q4 GPTQ-W4	788	6.3	4.3	10.7	$+2.0$ $[-0.3, +4.3]$
BFCL q4 AWQ-style W4	788	5.7	9.5	15.2	-3.8 $[-6.5, -1.3]$
BFCL q8 GPTQ-W4	788	2.9	2.3	5.2	$+0.6$ $[-0.9, +2.3]$
BFCL q8 AWQ-style W4	788	4.1	2.7	6.7	$+1.4$ $[-0.3, +3.3]$
BFCL W4 pool					$+0.1$ $[-1.0, +1.2]$
ALFWorld q8 GPTQ-W4	134	9.0	12.7	21.6	-3.7 $[-11.2, +4.5]$
ALFWorld q8 AWQ-style W4	134	10.4	14.2	24.6	-3.7 $[-11.9, +4.5]$
ALFWorld W4 pool					-3.7 $[-9.7, +1.9]$
BFCL q4 GPTQ-W3^s	591	6.1	3.6	9.6	$+2.5$ $[+0.2, +5.1]$
BFCL q8 GPTQ-W3^s	788	5.2	4.1	9.3	$+1.1$ $[-1.0, +3.3]$
BFCL q4 KV4^s	788	0.3	0.0	0.3	$+0.3$ $[+0.0, +0.6]$
BFCL q8 KV4^s	788	0.3	0.0	0.3	$+0.3$ $[+0.0, +0.6]$
ALFWorld q8 GPTQ-W3^s	134	6.7	3.0	9.7	$+3.7$ $[-1.5, +9.7]$
ALFWorld q8 KV4^s	134	0.0	0.0	0.0	$+0.0$ $[+0.0, +0.0]$

B. First divergent states

Cell	states	G	H	F	$G - H$	off-diag. %
BFCL q4 GPTQ-W4	573	$+2.8$ $[-0.4, +5.9]$	$+1.4$ $[-0.7, +3.5]$	$+4.2$ $[+0.9, +7.2]$	$+1.4$ $[-1.8, +4.5]$	11.7
BFCL q4 AWQ-style W4	642	-4.7 $[-7.8, -1.5]$	-1.1 $[-3.6, +1.4]$	-1.4 $[-4.8, +1.7]$	-3.6 $[-6.7, -0.2]$	17.1
BFCL q8 GPTQ-W4	533	$+0.9$ $[-1.3, +3.3]$	$+1.5$ $[+0.0, +3.2]$	$+2.1$ $[-0.4, +4.6]$	-0.6 $[-3.1, +2.0]$	4.7
BFCL q8 AWQ-style W4	548	$+2.0$ $[-0.4, +4.7]$	$+1.8$ $[+0.0, +3.7]$	$+3.6$ $[+0.9, +6.5]$	$+0.2$ $[-2.6, +3.1]$	7.8
BFCL W4 pool		$+0.3$ $[-1.2, +1.8]$	$+0.9$ $[-0.2, +2.0]$	$+2.1$ $[+0.3, +3.9]$	-0.6 $[-2.3, +1.0]$	10.3
ALFWorld q8 GPTQ-W4	115	-4.3 $[-13.3, +5.0]$	$+7.0$ $[-2.6, +16.7]$	$+9.6$ $[-0.9, +19.7]$	-11.3 $[-20.9, -1.8]$	40.0
ALFWorld q8 AWQ-style W4	119	-4.2 $[-13.6, +4.9]$	$+5.0$ $[-2.6, +12.2]$	$+4.2$ $[-5.9, +13.7]$	-9.2 $[-18.8, +0.0]$	32.8
ALFWorld W4 pool		-4.3 $[-11.1, +2.2]$	$+6.0$ $[-0.5, +12.6]$	$+6.9$ $[-2.1, +15.5]$	-10.3 $[-17.7, -2.9]$	36.4
BFCL q4 GPTQ-W3^s	540	$+2.8$ $[+0.2, +5.5]$	$+5.2$ $[+2.9, +7.7]$	$+8.9$ $[+5.7, +12.2]$	-2.4 $[-5.7, +0.7]$	13.0
BFCL q8 GPTQ-W3^s	696	$+1.3$ $[-1.1, +3.7]$	$+1.4$ $[-0.1, +2.9]$	$+3.4$ $[+0.9, +6.0]$	-0.1 $[-2.6, +2.4]$	8.3
BFCL q4 KV4^s	787	$+0.3$ $[+0.0, +0.6]$	$+2.9$ $[+1.9, +4.1]$	$+20.3$ $[+17.6, +23.0]$	-2.7 $[-3.9, -1.5]$	20.5
BFCL q8 KV4^s	788	$+0.3$ $[+0.0, +0.6]$	$+1.8$ $[+1.0, +2.7]$	$+14.5$ $[+12.1, +17.0]$	-1.5 $[-2.5, -0.6]$	14.7
ALFWorld q8 GPTQ-W3^s	130	$+3.8$ $[-1.5, +9.8]$	$+38.5$ $[+29.8, +46.9]$	$+42.3$ $[+33.6, +50.8]$	-34.6 $[-43.5, -25.4]$	30.8
ALFWorld q8 KV4^s	134	$+0.0$ $[+0.0, +0.0]$	$+38.1$ $[+30.6, +45.5]$	$+53.0$ $[+44.8, +61.2]$	-38.1 $[-45.5, -30.6]$	26.9

Table 18: Replacement and four-branch estimands on BFCL multi-turn and ALFWorld (no LLM user simulator; one run per item; 95% item-bootstrap intervals). All items: rescue r, damage d, turnover M (% of all items) and net G_{all} (pp), non-divergent items counting as zero. Divergent items (supplementary, exploratory, post hoc; pp, divergent items): Off-diag.: share with $\Delta_{\text{Q}} \neq \Delta_{\text{B}}$. Pools: equal weight per cell. ^{s}Stress and unrotated KV4 cells; in unrotated KV4, $Y_{\text{QQ}} = 0$ on every state, so positive gains are rescues; G, H and F remain distinct.

Cell	blocks	states	G	H	F
AWQ-style W4	3	255	$+2.7$ $[-4.6, +10.1]$ $[-2.4, +9.0]$	$+4.3$ $[-0.1, +8.8]$ $[-1.2, +11.0]$	$+5.5$ $[+1.0, +10.0]$ $[-0.8, +12.5]$
GPTQ-W4	3	255	$+0.4$ $[-7.0, +7.7]$ $[-4.3, +5.9]$	$+0.8$ $[-5.3, +6.9]$ $[-5.9, +7.5]$	$+3.5$ $[-2.3, +9.4]$ $[-3.9, +10.6]$
KIVI-16 path	3	255	$+0.0$ $[-2.9, +2.9]$ $[-2.7, +2.7]$	$+1.2$ $[-6.6, +8.9]$ $[-3.5, +5.9]$	$+2.4$ $[-0.6, +5.3]$ $[-2.4, +7.1]$
KIVI-2	2	170	-2.9 $[-25.4, +19.5]$ $[-8.2, +2.4]$	$+2.9$ $[-4.5, +10.4]$ $[-3.5, +10.0]$	$+2.9$ $[-34.4, +40.3]$ $[-5.9, +10.6]$
KIVI-4	3	255	$+0.0$ $[-5.1, +5.1]$ $[-3.5, +4.3]$	$+2.0$ $[-1.4, +5.3]$ $[-3.5, +7.5]$	$+0.8$ $[-8.1, +9.7]$ $[-5.9, +7.1]$
KV8	3	255	$+1.6$ $[-0.1, +3.3]$ $[-1.2, +4.7]$	$+5.1$ $[+0.6, +9.6]$ $[+0.4, +9.8]$	$+5.5$ $[+1.0, +10.0]$ $[+0.0, +10.6]$
QuaRot-W16 path	3	255	$+0.0$ $[-5.1, +5.1]$ $[-3.5, +3.1]$	$+1.6$ $[-14.5, +17.7]$ $[-5.9, +9.4]$	$+2.4$ $[-8.2, +12.9]$ $[-4.3, +9.0]$
QuaRot-W4	3	255	$+0.8$ $[-5.3, +6.9]$ $[-3.5, +5.9]$	$+1.6$ $[-1.8, +4.9]$ $[-4.3, +7.5]$	$+4.7$ $[+1.8, +7.6]$ $[-2.0, +11.4]$
QuaRot-W4A4	3	255	$+1.2$ $[-6.6, +8.9]$ $[-4.7, +6.3]$	$+6.3$ $[-8.4, +21.0]$ $[-2.4, +14.1]$	$+9.4$ $[+1.7, +17.1]$ $[+2.0, +16.5]$
SpinQuant-W16 path	3	255	-0.4 $[-2.1, +1.3]$ $[-3.1, +2.4]$	$+2.0$ $[-1.4, +5.3]$ $[-2.4, +6.3]$	$+2.4$ $[-0.6, +5.3]$ $[-2.4, +7.1]$
SpinQuant-W4A4	3	255	$+3.1$ $[-2.9, +9.2]$ $[-2.0, +8.2]$	$+11.8$ $[+6.7, +16.8]$ $[+5.1, +18.4]$	$+13.7$ $[+9.3, +18.2]$ $[+5.5, +21.6]$

Table 19: All q8 ordinary BF16-sourced states admitted by the panel rules, including unquantized path controls. Effects in pp; each cell: estimate with fixed-pool 95% t interval, then crossed task$\times$block 95% interval. All three modes use the same admitted states within a row. These descriptive mode intervals are unadjusted. KIVI-2 admits only two blocks after one-time infrastructure re-runs; it is not estimable under the original strict rule. Other cells admit three blocks. No action-only interval excludes zero; several takeover or joint-upgrade intervals do.